



TRAMONTANA ICS-01 – THE DISTRIBUTION OF IMPROVED COOKSTOVES TO INDIGENOUS COMMUNITIES ACROSS RURAL INDIA

BY

TRAMONTANA

Document Prepared by

Tramontana Asset Management

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Background

A recent Government of India report for 2019-21¹, concluded that 43.3% of the Indian population use solid fuels for cooking. The majority of these, or 33.4% of the total population which equates to approximately 450 million people, specifically use wood and are located in rural areas. Cookstoves form a central component of a family's life in rural communities, as they are typically used for cooking, room heating and boiling water for drinking and bathing. The traditional stove type used is predominantly a fixed mud clay stove (shown in Figure 1 and Figure 15) and in some cases a three-stone stove (shown in Figure 3). The report¹ estimates that of the Indian population using solid fuels, 95.4% use mud clay stoves. The traditional stoves used are mainly fuelled by unsustainably sourced firewood from local forests. The firewood then often needs to be carried over long distances by headload, as depicted in Figure 1.



Figure 1. Indigenous community member in the project area using a traditional mud clay cookstove and another carrying firewood by headload to her home

Health Implications

The World Health Organisation estimated that over 800,000 people² in India lost their lives prematurely in 2019 due to household air pollution. This is the largest number of any country. The primary cause is the use of poorly designed traditional inefficient biomass cookstoves in homes, which results in incomplete combustion and large quantities of toxic pollutants being emitted into the immediate environment, in the form of smoke, soot and other particulate matter (PM).

¹ http://rchiips.org/nfhs/NFHS-5Reports/NFHS-5_India_Report.pdf

² <https://www.who.int/data/gho/data/indicators/indicator-details/GHO/household-air-pollution-attributable-deaths>

Environmental Implications

The World Bank³ estimates that traditional cookstoves have a low thermal efficiency, in the range 8-15% depending on their method of use. This results in significant demand for firewood, which causes large quantities of greenhouse gas (GHG) emissions and widespread deforestation and degradation of local forested areas.

Social Implications

Typical stove related tasks such as sourcing firewood, cooking and cleaning are predominantly undertaken by women. Moreover, the considerable time they spend collecting and carrying heavy bundles of firewood over long distances has resulted in physical health problems and exposure to dangers, such as wild animal attacks and sexual assault.

Economic Implications

Stove related tasks in total consume between an estimated 6 and 8 hours per day on average (refer to section 3.4). It also forms the primary role of female members in the communities. This minimises the time they have for undertaking more productive activities such as paid employment, agricultural work or spending time in education or training.

Improving The Situation

The Tramontana ICS-01 project, certified under the Verra Verified Carbon Standard (VCS), aims to use carbon finance to distribute up to 400,000 safe energy-efficient heavy-duty improved cookstoves (ICS). The ICS, which are manufactured locally in India, will be provided free of cost to households across multiple states of the country. The project specifically targets low income and socially disadvantaged indigenous communities, who are generally considered to be the most marginalised and underrepresented in Indian society. These communities have been unable to afford or access cleaner cookstoves primarily due to their economic circumstances and remote locations. The project aims to directly and immediately enhance the health, well-being and livelihoods of the communities, and particularly of women and children who are the most affected.

Implementation Plan

Tramontana has identified multiple target areas across several states of India which suffer from local deforestation and are primarily populated with indigenous communities. A high proportion of the families in the project area own a Below Poverty Line (BPL) or ration support card issued by the government. This card indicates that the whole family has a maximum annual income of Rs 60,000 (or equivalent to approximately USD2 per day) (see section 1.14).

A distribution plan covering multiple instances under a group project has been developed. The initial instance is located in the state of Andhra Pradesh, from which the project can expand to other states including Odisha, Telangana and Chhattisgarh. The first project instance will distribute

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<https://documents1.worldbank.org/curated/en/600391468771661997/pdf/ESM2760PAPER0Impact0of0Energy.pdf>

approximately 50,000 ICS after which multiple instances are planned according to the schedule presented in Table 1. Further information on the project location is provided in section 1.13.

The total GHG emissions reductions estimated from the project activity over the first crediting period is 9,171,123 tonnes of CO₂ equivalent (tCO₂e), with an annual average of 1,310,160 tCO₂e (see section 5.4).

Implementation Status

Over the first monitoring period between 21-January-2023 and 23-August-2023, 46,160 ICS were installed in households. Information on ICS installation and the beneficiaries is recorded in the distribution database, which is managed by the Project Proponent.



Figure 2. Tramontana site visit to oversee ICS distribution and a recipient completing an ICS survey via the Tramontana ICS software application

Table 1. Planned distribution schedule of ICS across the project area corresponding to instances under the grouped project

Instance	Planned ICS Distribution
Region 1	50,000
Region 2	100,000
Region 3	100,000
Region 4	100,000
Region 5	50,000

Table 2. Ex-post emissions reduction table for the first monitoring period

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU's (tCO ₂ e)	Total VCU's (tCO ₂ e)
21-January-2023 to 23-August-2023	--	--	--	62,743	62,743
Total	--	--	--	62,743	62,743

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Joint Validation + Verification</i>	<i>21-January-2023 – 23-August-2023</i>	VCS	<i>Earthood Services Private Limited</i>	0.586

1.3 Sectoral Scope and Project Type

Sectoral scope ⁴	03 – Energy Demand
Project activity type	Project Type: II – Energy Efficiency Improvement Project Grouped Project: Yes

1.4 Project Eligibility

1.4.1 General Eligibility

The project involves energy-efficient cookstove distribution which falls under the sectoral scope 03 - *Energy demand* and is supported by a methodology approved under the VCS Programme. The project activity does not fall under any of the categories of excluded projects stated in the VCS Standard Table 2.1. Therefore it is eligible under the scope of the VCS Programme.

The project has followed the most recent version of the VMR0006 methodology version 1.2 published in July 2023. It was successfully listed in the Verra registry on the 16th June 2023. Note that since listing, the document has been updated from the previous version of the methodology.

⁴ <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

In line with section 3.6.8 of the VCS Standard v.4.7⁵ which states that “the project proponent shall include in a singular project all project activity instances within ten kilometres of another instance of the same project activity and with the same project proponent”, all project activity instances included to date and in the future within close proximity (ten kilometres) of one another will form part of this single project, and will not be separated into a new project.

1.5 Project Design

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ **The project is a grouped project**

Grouped Project Design

For the inclusion of new project activity instances, the Project Proponent must ensure that it meets the following eligibility criteria from the Verra VCS Standard, version 4.7⁵

Eligibility criteria	Compliance by new project instances
Meet the applicability conditions set out in the methodology applied to the project	All new instances under the grouped project will meet the conditions of the methodology as specified in section 3.2
Use the technologies or measures specified in the project description	The technology type employed is an energy efficient ICS to replace the traditional inefficient cookstoves used by households. Information on the ICS technology used for all instances is provided in the form of manufacturer product specifications, independent third-party certification for ICS thermal efficiency and through distribution data/photos.
Apply the technologies or measures in the same manner as specified in the project description	Only thermally efficient ICS will be distributed to households under project instances, and no other technologies will be adopted

⁵ <https://verra.org/wp-content/uploads/2024/04/VCS-Standard-v4.7-FINAL-4.15.24.pdf>

Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area	All new project instances will remain within the geographical boundaries of India and will therefore share some similar characteristics. However, given each instance corresponds to a different region, separate baseline campaigns are conducted to ensure a more accurate representation of the respective region. The baseline scenario, including for the first instance, is discussed in section 3.4. Baseline data for subsequent instances is provided during verification.
Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area	All new project instances will have additionality characteristics consistent with the initial instance, and this will be demonstrated using the activity methodology below. Step 1: Regulatory Surplus Any new project instances included will not be mandated by any law, government policy or regulation in the host country Step 2: Positive List The inclusion of new project instances will comply with the positive list under the methodology (see section 3.5 for details) Step 3: Project Method Not applicable

1.6 Project Proponent

Organisation Name	Tramontana Impact Limited (subsidiary of Tramontana Asset Management)
Contact person	Dr Ramanan M. Loganathan
Title	Vice President
Address	2nd Floor, Gaspé House, 66-72 Esplanade, St. Helier, Jersey, JE1 1GH
Telephone	+44 (0)203 773 0160

Email	impact@tramontana.co.uk
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1.7 Other Entities Involved in the Project

Organisation Name	Cropzone Agro Forestry Private Limited (subsidiary of Tramontana Asset Management)
Role in the project	Local Implementation
Contact person	Mr Saibaba Abburi
Title	Managing Director
Address	1130/A, 3rd Floor, Usha Jubilee Tower 36, Road No. 36, Jubilee Hills, Hyderabad, Telangana-500033, India
Telephone	+91 98 4960 2999
Email	saibaba@cropzone.in

1.8 Ownership

The project ownership is with the Project Proponent. All rights to carbon credits (or Verified Carbon Units under Verra) generated from project instances will be with the Project Proponent by means of a standardised licence agreement physically signed by the beneficiary of each ICS at the ICS distribution stage.

1.9 Project Start Date

Project start date	21-January-2023
Justification	Date of installation of the first ICS under the grouped project

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
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Start and end date of first or fixed crediting period

21-January-2023 to 20-January-2030

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☐ < 300,000 tCO₂e/year (project)

☒ ≥ 300,000 tCO₂e/year (large project)

The following are annual emissions reduction estimates from the project activity over the first crediting period. Details of the assumptions and calculations are provided in section 5.

Calendar Year of Crediting Period	Estimated GHG emissions reductions (tCO ₂ e)
21-Jan-2023 to 31-Dec-2023	206,521
01-Jan-2024 to 31-Dec 2024	860,231
01-Jan-2025 to 31-Dec 2025	1,646,313
01-Jan-2026 to 31-Dec 2026	1,719,647
01-Jan-2027 to 31-Dec 2027	1,641,975
01-Jan-2028 to 31-Dec 2028	1,555,844
01-Jan-2029 to 31-Dec 2029	1,459,791
01-Jan-2030 to 20-Jan-2030	80,801
Total estimated ERs (first crediting period)	9,171,123
Total number of years	7
Average annual ERs	1,310,160

1.12 Description of the Project Activity

Household air pollution from traditional inefficient biomass cookstoves in rural homes is a leading cause of premature deaths in India (see section 1.1). Despite recent government efforts to promote cleaner cooking fuels (through schemes such as PMUY), challenges have occurred including uptake owing to high fuel costs (e.g. for LPG refills) and supply chain issues with fuel distribution to rural areas. The project activity enables the distribution of Tramontana ICS (Figure 4) directly to individual households in rural areas across multiple states in India. These ICS are provided free of cost, using carbon financing as the means to overcome a key hurdle for their adoption. They will replace traditional inefficient cookstoves, typically fixed mud clay stoves (made using mud, dung and brick or stone) (see Figure 1 and Figure 15) or three-stone stoves (see Figure 3), which use woody biomass as a fuel and have been utilised by the communities for generations.



Figure 3. A traditional three-stone cookstove and a community member in the project area using this stove type

The project is aimed primarily at households who are marginalised from a socio-economic standpoint. Many households belong to communities that have been statutorily recognised (in the Indian Constitution and Law) as Scheduled Castes and Scheduled Tribes (defined in Article 341 and 342 respectively of the Indian Constitution), which are considered the most disadvantaged socio-economic groups in India. The state government often provides documentation allowing these households to prove their community status, which in turn allows them to avail of several targeted welfare schemes such as subsidised food grains and preferential access to education and jobs. Note that evidence of such document is collected from ICS beneficiaries and stored electronically during the distribution phase of the project activity (see section 1.12.1).

The Tramontana ICS to be distributed forms a safe, portable cleaner cooking technology which can be (voluntarily) adopted by the communities and does not require them to make major modifications to their current cooking practices. The ICS has an increased thermal efficiency and completeness of combustion through its energy-efficient design, technology and the materials used. GHG emissions are immediately reduced, and families experience significant improvements in the levels of toxic indoor air pollution. The improved efficiency lowers deforestation/forest

degradation (and consequently biodiversity loss) and enhances the daily lives particularly of female family members, who are most involved in carrying out the daily tasks relating to the stove.

A pilot study was conducted starting in January 2022, approximately one year in advance of distribution of the first ICS under the project. The study involved distributing 100 ICS within the project region and continuously monitoring usage and performance. This enabled the Project Proponent to make necessary ICS design improvements and incorporate feedback from community members. The pilot study also helped validate the robustness of the entire supply chain.



Figure 4. Tramontana ICS with unique serial number, QR code and Tramontana branding

Upon project implementation, as part of the ICS distribution process, the project team directly engage with each household and complete a number of tasks including:

- Provide physical demonstrations on how to use the Tramontana ICS
- Make recommendations on best practices including with fuel preparation (e.g. bark shredding, firewood sizing) and stove maintenance
- Hand over the assigned ICS together with an instructions brochure
- Use the Tramontana ICS software application to record an extensive set of data on the user (as outlined in section 1.12.1)
- Undertake post distribution checks in the weeks following distribution to assess initial performance and identify any issues

The technical specifications of the Tramontana ICS distributed under project activity instances are outlined in Table 3 and Table 4. These specifications will be used to determine emissions reductions for the project. If at a later date, an additional ICS model is included in an instance or an improvement is made to the existing ICS, their specifications will be provided during verification and correspondingly accounted for in emissions reduction calculations.

Table 3. Technical specifications of the Tramontana ICS distributed in the project area

ICS Model	CRS-23
Type	Natural Draft
Thermal Efficiency	36.01%
Height	25.5 cm
Diameter	27 cm
Weight	9.471 kg
Eligible Fuel Type	Wood, Charcoal, Other Biomass
Portable or Fixed	Portable
Colour / Finish	Black & White / Glossy
Expected Lifespan	7 years

Table 4. Material components of the Tramontana ICS distributed in the project area

ICS Part	Material	Thickness	Tolerance
Top Plate	Cold Rolled (CR) Steel	0.8 mm	+/- 0.1 mm
Grate	Cast Iron	6 mm	+/- 0.5 mm
Barrel/Combustion Chamber	Ceramic	23 - 65 mm (height variation)	+/- 2 mm
Bottom Ring	CR Steel	1 mm	+/- 0.02 mm
Neck (Outer Jacket)	CR Steel	0.6 mm	+/- 0.02 mm
Neck (Inner Lining)	Stainless Steel 430	0.6 mm	+/- 0.02 mm
Perforated Sleeve	CR Steel	0.6 mm	+/- 0.02 mm
Outer Jacket Cladding	CR Steel	0.6 mm	+/- 0.02 mm
Handles	Mild Steel wire covered with Bakelite	8 - 12 mm	+/- 2 mm

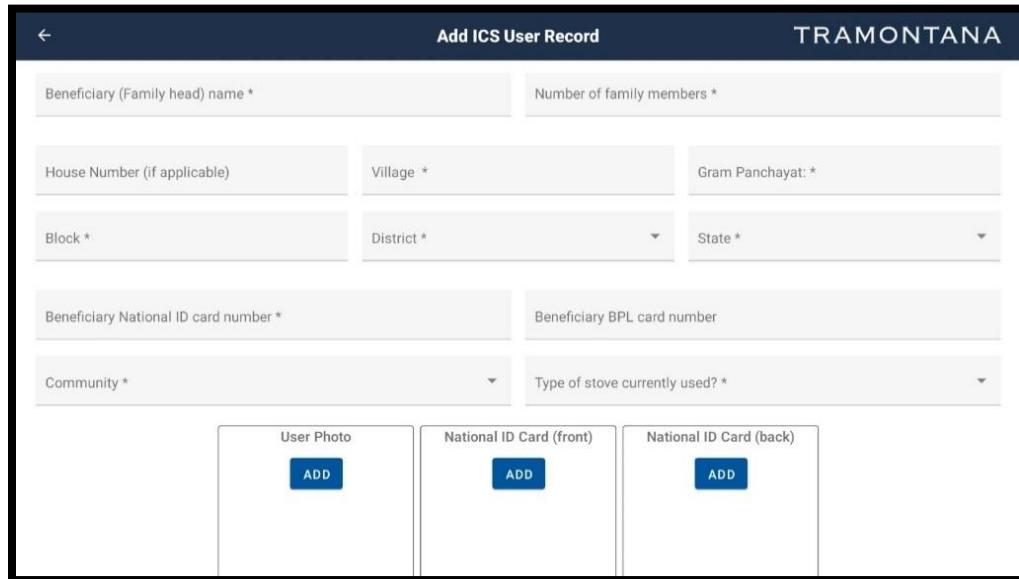
Pan Support	CR Steel	2 mm	+/- 0.02 mm
Legs	CR Steel	1 mm	+/- 0.02 mm
Fuel Stand	Mild Steel	4 mm	+/- 0.08 mm

1.12.1 Data collection from ICS end user

The Project Proponent will obtain the necessary information to identify households using its ICS during the course of the project. To facilitate this process, the Project Proponent has specifically developed the Tramontana ICS software application which is its own proprietary application on the Android operating system (OS) for conducting surveys. The application runs on portable tablets which are used by trained field staff specifically employed for the project. Each ICS (see details in Table 3 and Table 4) is assigned a unique serial code which is displayed on the device in writing and in the form of a QR code. For a new end user record as part of the registration process, the QR code is scanned using the Tramontana ICS software application and the following information is recorded. The information includes (as appropriate and as available):

- Name, phone number and photo of beneficiary/head of household with their ICS
- Address of ICS household
- Statutory Community Classification (if any)
- Pre-project cookstove type
- Photos of ID card and other certification
- ICS model
- ICS serial number
- GPS coordinates of ICS household
- Date and time of ICS installation
- Signed ICS licence agreement (including credit ownership)
- On-field distribution team member information including name, title and signature

Once a new user record is created in the Tramontana ICS software application it is automatically synced to a Google cloud database after the tablet is brought online. Figure 5 is a screenshot of a section of the application used for registration of new ICS beneficiaries. All information collected/recorded including for baseline, registration and monitoring purposes is stored in the online database and subsequently automatically mapped into an Excel sheet. This sheet will serve as the basis for monitoring and auditing throughout the project life.



The screenshot shows the 'Add ICS User Record' form in the TRAMONTANA application. The form is organized into several sections:

- Top Section:** Contains 'Beneficiary (Family head) name *' and 'Number of family members *'.
- Location Section:** Includes 'House Number (if applicable)', 'Village *', 'Gram Panchayat: *', 'Block *', 'District *' (with a dropdown arrow), and 'State *' (with a dropdown arrow).
- Identification Section:** Features 'Beneficiary National ID card number *' and 'Beneficiary BPL card number'.
- Community and Stove Section:** Includes 'Community *' (with a dropdown arrow) and 'Type of stove currently used? *' (with a dropdown arrow).
- Photo Upload Section:** At the bottom, there are three boxes for adding photos: 'User Photo', 'National ID Card (front)', and 'National ID Card (back)'. Each box contains an 'ADD' button.

Figure 5. Screenshot from the new ICS user record section of the Tramontana ICS software application

1.13 Project Location

All project activity under the grouped project is within the geographical boundaries of India depicted in Figure 6⁶ with approximate coordinates of latitude 21.555808 and longitude 78.746972. A separate KML file is provided describing the project location. The first project instance is within the Alluri Sitarama Raju (ASR) district of the state of Andhra Pradesh shown in Figure 7⁷, with approximate coordinates of latitude 18.011959 and longitude 82.555030. The elevation of this region is approximately 1000 metres.



Figure 6. Project activity area of India delineated by state

⁶ <https://www.mapsofindia.com/images2/india-map.jpg>

⁷ <https://www.india.com/news/india/andhra-pradesh-to-have-13-new-districts-from-april-4-check-list-of-total-26-districts-here-5317197>



Figure 7. Map of Andhra Pradesh state in India divided by district where the first project activity instance is located (see blue arrow)

1.14 Conditions Prior to Project Initiation

The target communities live in rural areas and are predominantly from indigenous and socio-economically disadvantaged communities. They rely on nature for their livelihoods, and reside in environments ranging from forests to hills with many in difficult-to-access remote areas. The level of education is generally low and many children lack access to schools or drop out early to help out their family. Most families own a Below Poverty Line (BPL)/ration card issued by the state government confirming eligibility to receive food provisions, subsidies etc, which in Andhra Pradesh (and similarly in other states) is only handed to those households with a maximum annual income for the whole family of less than Rs 60,000 (or equivalently USD2 per day approximately) amongst other conditions⁸. Documentary proof of this, where available, is collected during ICS distribution and recorded in the project distribution database.

The pre-project baseline scenario is the continued use of rudimentary traditional cookstoves by the communities, typically in the form of fixed mud clay stoves (Figure 1 and Figure 15) or three-stone stoves (Figure 3). Usage of these stoves has been commonplace in the project areas for generations, where households burn woody biomass to meet their thermal energy needs including for the purposes of cooking, heating and boiling water. These cookstoves are considered to have a low thermal efficiency, in the range 8-15% depending on their method of use according to the World Bank⁹. Further, their poor design results in incomplete combustion which releases significant quantities of toxic air pollutants into the immediate environment. This causes issues such as breathlessness, coughing, headaches, burning eyes and longer term respiratory/retinal issues in

⁸ <https://www.civilsupplies.ap.gov.in/AnnIV-Iss-BPL.jsp>

⁹

<https://documents1.worldbank.org/curated/en/600391468771661997/pdf/ESM2760PAPER0Impact0of0Energy.pdf>

the target communities, particularly as users (mainly women) spending long hours in close proximity to the stoves, and in many cases within a confined environment.

The large firewood requirement of traditional stoves results in significant generation of GHG emissions and deforestation/degradation of the local forested areas (see Figure 8). It further leads to considerable time spent collecting firewood, which prevents primarily female family members from undertaking alternative income generating and skill development activities. Given the heavy loads are carried by headload and for many kilometres, it also causes physical health problems as well as exposure to dangers from the local area including sexual assault and wild animal attacks. To gain further insight into the pre-project conditions, baseline surveys are conducted for each project activity instance under the grouped project. Details for the first project instance can be found in section 3.4.



Figure 8: Community member chopping trees to collect firewood to carry home and evidence of recent deforestation in the local area as result of large firewood demand

Under the project activity, all beneficiaries will communicate information on their pre-project stove when receiving a new ICS, with responses recorded in the distribution database. There will also be a requirement for demolishing or removing pre-project cookstoves prior to receiving a new ICS in order to minimise any chances of reverting to the previous stove.

1.15 Compliance with Laws, Statutes, and Other Regulatory Frameworks

There exists no mandatory law or regulations requiring the introduction or use of ICS in India. The project is undertaken on a voluntary basis by the Project Proponent.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes

☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS Program Definitions for definitions of emissions trading program and binding emissions limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The project contributes to environmental, social, economic and technological benefits which lead to the sustainable development of the local area and country as follows:

Environmental Benefits

- Preservation of forest stock and protection of ecosystems, habitats and soil health by reducing the rate of deforestation and forest degradation across the project area.
- Reduction in toxic indoor air pollution from the cleaner combustion characteristics and increased thermal efficiency of the project ICS. This lowers health issues from respiratory and other diseases. Communities also save on medical expenditure and require less time off to look after unwell family members.
- Lower GHG emissions generated by a reduction in the use of non-renewable biomass.
- Less water and effort needed for cleaning utensils and maintenance of the household as a result of the cleaner cooking process and minimised smoke/soot generation.

Social Benefits

- Reduced drudgery and exposure to risks including wild animal attacks, sexual assault, injuries from tree cutting/wood chopping etc. that are prevalent in the project area, as a result of less time needing to be spent sourcing, carrying and processing firewood. These issues were widely reported in baseline surveys for the first project instance (see section 3.4).
- More hours for skill/economic development and schooling owing to reduced time spent sourcing firewood.
- Improvement to gender equality as it is primarily women that carry out the tasks relating to stove usage, firewood collection and cleaning.
- Improved home conditions and quality of life for families by reducing localised indoor air pollution and from the time savings.
- Enhanced safety and reduced burns/injuries for children owing to the compact nature of the Tramontana ICS and less direct exposure to the flame.
- Confidence and comfort as no material change is required to traditional cooking practices.
- Development of IT skills for the local teams employed as a result of training and experience gained under the project activity.

Economic Benefits

- Employment opportunities and skill development for the local communities and economy through the project activity. This includes specifically employing and training local teams for operational tasks such as distribution, monitoring, maintenance, data analysis and conducting stakeholder meetings.
- More hours available for productive income generating activities and schooling as a result of reduced household work and fuel sourcing.

- Reduced expenditure on firewood (if purchased) due to the increased thermal efficiency of the Tramontana ICS.

Technological Benefits

- Introduction of cleaner technology directly to rural communities.
- Employment of local workers and provision of training and experience in the use of tablets with an Android software application to conduct baseline surveys, distribution and monitoring work.
- Provision of training and experience in IT for local office teams to undertake the tasks around project data compilation and analysis.

1.18.2 Sustainable Development Contributions Activity Monitoring

Nine of seventeen global UN Sustainable Development Goals (SDGs) have been identified as relevant to this project and are illustrated with black boxes below. Table 5 provides the detailed description of each of these SDGs including the relevant baseline scenario and corresponding contribution from the project activity. Information is also provided on how each SDG contribution will be measured through the project life.



During the project's first monitoring period up to 23rd August 2023, 46,160 portable ICS devices were distributed to households in Andhra Pradesh. The impact of the project activities during the period have been monitored using the monitoring framework detailed in section 6. Table 6 presents the project's contributions to the specific SDGs outlined in Table 5, which were recorded during monitoring. Note in Table 6 that current project contributions are the same as the cumulative over the lifetime as this is the first monitoring period.

Table 5. Information on relevant SDGs for the project including outlining the baseline scenario as well as the corresponding contribution from the project activity and monitoring approach used

SDG Target	SDG Indicator ¹⁰	Baseline Scenario	Net Impact on Indicators	Contribution of Project Activity and Monitoring Approach
 NO POVERTY	1.2.1 Proportion of population living below the national poverty line, by sex and age 1.4.1 Proportion of the population living in households with access to basic services	The majority of the households in the target communities belong to those that have been statutorily recognised (in the Indian Constitution and Law) as coming from the most disadvantaged socio-economic groups in India. The communities typically own a Below Poverty Line (BPL)/ration government card signifying a maximum annual family income of Rs 60,000 or equivalently USD2 per day approximately (see section 1.14). The communities are currently using rudimentary inefficient and highly polluting traditional stoves to meet their thermal energy needs. They are unable to purchase an ICS primarily due to lack of access and affordability, and thus live without the means	Implemented activities to decrease 1.2.1 Implemented activities to increase 1.4.1	Under the project activity, ICS will be distributed to rural communities that are considered the most disadvantaged socio-economic groups in India. The ICS will provide them with access to clean cooking technology which is considered a basic service for all families/households. In addition, it also reduces the time spent by families in undertaking stove related tasks and therefore allows more time for: income generating activities (e.g. supporting in the main work contributing to the family's income), activities to help with skill development, and community work, which can all improve the family's livelihood. The impact on this SDG can be assessed by first identifying the number of households owning a BPL/ration card through survey responses and documentary proof (where available) in the distribution database. Then subsequently, the number of households from these community categories that have received and are using an ICS, and the

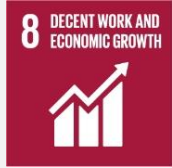
¹⁰ www.sdgdata.gov.uk

		to access basic services including clean cooking.		benefits of time savings, based on distribution and monitoring database records.
 GOOD HEALTH & WELL-BEING	3.9.1 Mortality rate attributed to household and ambient air pollution	The poor design of traditional stoves used in the target communities results in incomplete combustion of the firewood which releases significant toxic air pollution into the local environment as smoke, soot and other PM. This has contributed to a prevalence of respiratory and retinal issues, and premature deaths in the local communities as users (mainly women) spend long hours in close proximity to the stoves. Further observations on the baseline scenario are given in sections 1.14 and 3.4.	Implemented activities to decrease	The improved quality and design of the Tramontana ICS results in an immediate improvement in the levels of smoke/soot generation and toxic indoor air pollution, therefore contributing to improved home conditions for families. Additionally, the portable nature of the ICS provides flexibility in the location of use, which can be for instance elsewhere together with the family or outside. The increased efficiency of the ICS means less time is spent sourcing and carrying firewood. This means less of the negative physical and social issues associated with these tasks, and more time with the family and for pursuing alternative tasks. The impact on this SDG can be assessed firstly by identifying the number of households using their traditional stoves in an enclosed space, and then through monitoring surveys obtaining extensive user feedback on areas including ICS usage/performance, changes in local conditions and impact on general well-being. The information is presented in the monitoring database and reports.

 QUALITY EDUCATION	4.3.1 Participation rate of youth and adults in formal and non-formal education and training in the previous 12 months, by sex 4.4.1 Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill	The socio-economically disadvantaged target communities have limited support, access or even time available to be able to undertake formal education or skill development/training. Instead, much of the time of family members (including children) is spent helping to carry out tasks for the purposes of survival, such as firewood gathering, cooking/heating, cleaning, agricultural labour for subsistence purposes etc.	Implemented activities to increase	Given the lack of access and opportunities available to community members and their children, improving the time available for education forms an important aspect to the project. The time savings from the reduced firewood collection, as a result of the improved design and efficiency of the ICS, leads to more time available for education. Employed staff are trained under the project to undertake tasks including surveying and data collection using the Tramontana application. Training includes IT skills (using the application) and methods to maintain high data quality/integrity. Each household that becomes a beneficiary of an ICS is educated on areas such as climate change/sustainability, sustainable stove usage practices, ICS design and benefits, firewood preparation, maintenance/repair etc and is provided with continuous support and engagement throughout the project life. The impact on this SDG can be assessed through monitoring survey results identifying the number of beneficiaries (predominantly children) that are able to spend more time in education, as well as, the number of staff employed under the project and given training for their role.
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 GENDER EQUALITY	5.2.2 Proportion of women and girls aged 15 years and older subjected to sexual violence by persons other than an intimate partner in the previous 12 months, by age and place of occurrence 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	It is primarily female members of families that spend a disproportionate amount of time carrying out the daily tasks (unpaid work) around stove preparation and usage. This makes them the most exposed to the negative aspects of stove related tasks including pollution, physical pain and social issues. It also prevents female family members from being able to undertake other activities that provide skill development, generate income or support the local community.	Implemented activities to decrease	Though the ICS project activity alone is unable to alter/balance the role of family members in carrying out the daily tasks, it can drastically reduce the negative impacts associated with firewood collection/preparation and stove usage which is undertaken mainly by the female members. This includes issues such as sexual assault which is commonly experienced during collection of firewood. The reduced negative impact is achieved as result of the improved quality, design and efficiency of the Tramontana ICS, and via the education provided on sustainable practices. The impact on this SDG can be assessed by first determining from the baseline database, which family members (particularly female) are carrying out stove related tasks as well as any issues that are faced during firewood collection, and then from monitoring surveys understanding whether beneficiaries are actually experiencing environmental, social and economic benefits. Survey responses focussed on here include changes experienced in the time to collect firewood, and any improvements observed in user health/well-being. In addition, the SDG impact is also assessed through the number of female members
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				recorded as employed and trained as part of the project activity.
 AFFORDABLE AND CLEAN ENERGY	7.1.2 Proportion of the population with primary reliance on clean fuels and technology 7.a.1 International financial flows to developing countries in support of clean energy research and development and renewable energy production, including in hybrid systems	There are approximately 600 million people in India that still have a continued reliance on solid fuels (mainly firewood) as a cooking fuel (see section 1.12) and do not have access to cleaner energy sources. Across the project area usage of rudimentary traditional inefficient cookstoves by the communities has been commonplace for generations. Households under the project belong to communities that have been statutorily recognised as suffering from socio-economic deprivation, and have been unable to afford any cleaner forms of energy	Implemented activities to increase	The Project Proponent supporting this project is an international company based outside of the project region. Through their investment into distributing high quality, efficient ICS free of cost, the target communities can directly and immediately experience the benefits of using cleaner energy/technology and without the need to significantly change their daily practices. The free of cost nature is particularly advantageous for the poor marginalised communities, who would otherwise not be able to afford cleaner technologies. The impact on this SDG can be assessed by multiplying the number of households that are using an ICS by the household size, to determine the total number of people under the project that have switched and are benefiting from cleaner energy/technology. The household size is determined at the distribution stage and the number of stoves operating can be identified from monitoring survey results for the period.

 DECENT WORK AND ECONOMIC GROWTH	8.5.2 Unemployment rate, by sex, age and persons with disabilities	The marginalised target communities have limited access to skilled jobs and development opportunities given their remote locations, the low-quality of resources/infrastructure available to them, and the minimal outside support. Most people work in subsistence for their families in areas such as agriculture and have very little income, as confirmed by the Below Poverty Line (BPL)/ration card issued to them by the state government.	Implemented activities to decrease	The project activity has resulted in new jobs, training and skill development for local people. Specifically, local field and office staff have been employed and supported with extensive training, including: gaining experience using tablets and an Android software application, with data entry/acquisition and analysis, and in event set up and hosting for stakeholder meetings and community engagements. In addition, opportunities for coordinator roles have been given to more experienced individuals. The project promotes decent work conditions for all by ensuring continuous support, fair wages and benefits, training, safety, and opportunities for career development. This can lead to a strong positive impact on the livelihoods of local workers and their families, and greater social and economic development/growth in the communities. Outside of the local project areas, many more people are employed (both directly and indirectly) to support the project activity, including for manufacturing, transport and logistics etc All these service providers are located within the project boundary of India and thus the project activity contributes to national development and growth. The impact on this SDG can be assessed by observing the records of all persons
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				employed (full-time and part time) under the project activity. Employment information is also obtained from outside service providers where possible. In addition to employment data, other information is collected such as attendance records for all training sessions, number of tablets deployed, stakeholder event details including local speakers and organising staff.
 RESPONSIBLE CONSUMPTION AND PRODUCTION	12.2.2 Domestic material consumption, domestic material consumption per capita, and domestic material consumption per GDP	The unsustainable practices employed across the project area and low thermal efficiency of traditional baseline cookstoves (in the range 8-15% as discussed in section 1.14) has resulted in the overuse of non-renewable biomass, thereby depleting precious local natural resources including forests (see for instance Figure 8). As these practices are widespread across the project area, the negative impact on local resources is significant.	Implemented activities to decrease	The Tramontana ICS distributed across the project region has an improved design/durability and significantly higher thermal efficiency as compared with the baseline stove. This results in an immediate reduction in the consumption of firewood, estimated to amount to an approximate 60% reduction by mass. This sizeable reduction for all project households will also lead to less time needing to be spent carrying out stove-related tasks including firewood sourcing, carrying and preparation work. Local communities are also educated to adopt sustainable practices and maximise resource efficiency going forward. The impact on this SDG can be assessed by calculating the total mass of firewood saved through the project activity. This is determined by multiplying the firewood consumption reduction per stove over its

				operating life by the number of operating stoves observed from monitoring surveys.
 CLIMATE ACTION	13.2.2 Total greenhouse gas emissions per year	Rudimentary traditional cookstoves which are poorly designed and considered to have a low thermal efficiency are widely used across the project area. This results in significant GHG emissions being generated and incomplete combustion leading to localised toxic air pollution in the form of smoke, soot and other PM.	Implemented activities to decrease	The replacement of traditional cookstoves with the efficient Tramontana ICS through the project activity will result in an immediate reduction in GHG emissions (and localised air pollution). GHG emissions reductions will occur throughout the lifetime of the project. As an ex-ante estimate based on calculations in section 5, the annual GHG emissions reductions over the first crediting period is 1,310,160 tCO_{2e}. The impact on this SDG can be assessed by calculating and verifying the emissions reductions that occur from the operational ICS in each monitoring period. As this is the primary task behind the generation of carbon credits (or Verified Carbon Units under Verra), the SDG does not require any additional monitoring to that already undertaken as part of monitoring/verification under the project activity.
 LIFE ON LAND	15.1.1 Forest area as a proportion of total land area	The firewood used by cookstoves across the project area is typically sourced by family members directly from natural forested areas (3.31km away on	Implemented activities to increase 15.1.1 and 15.2.1	The reduction in firewood consumption by the Tramontana ICS as a result of its higher thermal efficiency, leads to a meaningful increase in the preservation of forest stock, associated ecosystems, wildlife habitats and

	15.2.1 Progress toward sustainable forest management 15.3.1 Proportion of land that is degraded over total land area	average in the first project instance as indicated in section 3.4). It is obtained by collecting fallen wood or cutting down trees in the forest, sorting the wood and then carrying it home by headload and processing it. The firewood obtained is non-renewable (see section 3.2), thereby contributing to deforestation and forest degradation across the region (as shown in Figure 8). This has been occurring over generations with little effort to replenish the land through sustainable activities such as replanting.	Implemented activities to decrease 15.3.1	soil health through the reduced rate of deforestation and forest degradation. Local communities are also educated in stakeholder meetings and various community engagements on sustainable practices and the importance of its adoption to minimise critical forest loss. The impact on this SDG can be assessed by calculating the total mass of firewood saved through the project activity and subsequently estimating the amount of forested area preserved using average tree dimensions and spacings. As with SDG 12, the mass of firewood saved corresponds to the product of the firewood consumption reduction per stove over the period of operation and number of operating stoves obtained from monitoring survey results.
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Table 6. Information on relevant SDG contributions including stating the specific quantitative observations made during the first monitoring period, based on the criteria outlined in Table 5¹¹

SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions over Project Lifetime
 NO POVERTY	1.2.1 Proportion of population living below the national poverty line, by sex and age 1.4.1 Proportion of the population living in households with access to basic services	Implemented activities to decrease 1.2.1 Implemented activities to increase 1.4.1	Through the provision of ICS to the communities, direct access has been given to the basic service of cleaner cooking. Over the monitoring period a total of 46,160 ICS were distributed to beneficiaries. 100% of beneficiaries had a Below Poverty Line (BPL)/ration card issued by the state government These cards are only given to the most marginalised and underrepresented communities in India, with one of the conditions being a maximum family income of approximately USD2 per day. Each card has been photographed and recorded for every beneficiary in the distribution database. It was observed from the baseline surveys that the: - average annual income of project users was INR 25,091 or equivalently USD 0.84 per day, highlighting the poverty levels in the region. In relation to community status, 99.99% of beneficiaries confirmed belonging to the indigenous Scheduled Tribe/Adivasi or Scheduled Caste/Dalit communities (see section 1.12)	Same as current contributions

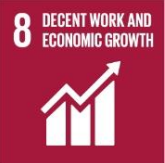
¹¹ All output values and calculations can be found in the baseline, distribution and monitoring databases provided alongside this document

			Where possible certification confirming a beneficiary's official community status was photographed during distribution (see distribution database). During monitoring, upon using their ICS: 100% of beneficiaries confirmed that they saved time in collecting firewood. With the time saved, 100% stated that they were able to spend more time supporting the main work contributing to the family's income 11% stated that they were able to undertake more community work These activities contribute to improving the livelihoods of the families and reducing poverty.	
 GOOD HEALTH & WELL-BEING	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	From baseline survey results, it was observed that 85.9% of respondents were using their traditional stove indoors in an enclosed space Further, in relation to smoke removal measures 48.7% had no provisions for smoke removal 33.2% had some small opening in the walls or ceiling 18% had a window - only 0.1% indicated that they had a chimney	Same as current contributions

			Upon using an ICS, of the monitored households: 100% stated that the local smoke pollution in their homes had decreased 100% confirmed experiencing general health improvements In addition, the portable nature of the ICS offers the flexibility to beneficiaries to use their ICS outside of their house. These benefits have resulted in communities experiencing less of the physical and mental health issues which were stated in the baseline database, and - in 96% of cases being able to spend more time with the family. During monitoring all beneficiaries stated that the function of their ICS was 'excellent'. In the comments section of the monitoring database, statements have been made by users on the improvements experienced to well-being and health, including reduced respiratory and retinal issues.	
 QUALITY EDUCATION	4.3.1 Participation rate of youth and adults in formal and non-formal education and training in the previous 12 months, by sex 4.4.1 Proportion of youth and adults	Implemented activities to increase	Improving the quality of education in the local communities is a key aspect of the project, particularly given the lack of access and opportunities available to the community members and their children. From the baseline database it was observed that 35% of children are currently involved in stove related tasks, thereby preventing them from focussing on education. Monitoring highlighted that with the time saved by using an ICS	Same as current contributions

	with information and communications technology (ICT) skills, by type of skill		66% of respondents indicated that their family (primarily children) were already able to spend more time in education. Throughout the ICS distribution phase and during all post-distribution engagements, information and support is provided on a continuous basis to all beneficiary households on the correct use of the ICS with proper fuel preparation, as well as educating them on the importance of employing sustainable practices. During monitoring, many beneficiaries also commented (see monitoring database) that they were aware of the local area project coordinators if ICS support was needed. All 302 employees under the project (see SDG 8) were given extensive training for their roles, involving use of tablets with the Tramontana application and methods to ensure high data quality/integrity. Those working on computers were given additional IT and data analysis training.	
 GENDER EQUALITY	5.2.2 Proportion of women and girls aged 15 years and older subjected to sexual violence by persons other than an intimate partner in the previous 12 months, by age and place of occurrence 5.4.1 Proportion of time spent on unpaid domestic	Implemented activities to decrease	Gender equality is a critical aspect of the project given the disproportionate number of female members who perform stove related tasks. It was observed from the baseline database that in: 97% of households the adult female family member is the primary stove user 78% of households women are relied upon to collect the firewood It was evident from the same database, that as a result of these tasks: 98.9% of end users experienced regular physical discomfort 99.9% had exposure to smoke and were aware of the dangers	Same as current contributions

	and care work, by sex, age and location		76.7% faced difficulties to obtain their firewood Many negative comments were also made, including users experiencing sexual harassment during collection of firewood. From monitoring records based on using their ICS over the monitoring period, 100% confirmed that their health and daily life had improved 100% stated that the local smoke pollution in their homes had decreased 100% experienced a reduction in the time to collect firewood All these improvements also minimise the impact of issues that were stated in the baseline database. By using the ICS it was found that family members, mainly women, were now able to: - spend more time contributing to the main family income source in 100% of cases - spend more time with the family in 96% of cases - in 11% of cases commence some form of community work Photos of female family members using their ICS, such as those presented in Figure 16, are recorded in the monitoring database. Across all the teams employed 20% were female. The proponent is seeking to hire and train more female members where possible as the project expands.	
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 AFFORDABLE AND CLEAN ENERGY	7.1.2 Proportion of the population with primary reliance on clean fuels and technology 7.a.1 International financial flows to developing countries in support of clean energy research and development and renewable energy production, including in hybrid systems	Implemented activities to increase	The carbon credit market has enabled international financial flows to contribute to cleaner energy adoption in a developing country by making such a project feasible. Over the monitoring period, it was observed that 100% of the households were currently using their ICS and had been using it over the monitored period Upon summing all family members across households that have received an ICS in the database, and factoring in an adoption factor for those that are using their ICS, it is calculated that cumulatively 162,543 people have direct access to cleaner energy/technology through the project. Given the very low average daily income of USD 0.84 (see SDG 1) identified from baseline surveys, providing ICS free of cost was found to be the only means to ensure mass adoption and project scalability. Continued ICS use is ensured through the high quality design of the ICS, continuous engagement with the communities, and education on the importance of sustainability.	Same as current contributions
 DECENT WORK AND ECONOMIC GROWTH	8.5.2 Unemployment rate, by sex, age and persons with disabilities	Implemented activities to decrease	The employment and training of people in country has resulted in skill development and improvements to the livelihoods of local people. This in turn contributes to local and national economic growth and development of the work force. Thus far the project has in total directly employed: 34 people full-time 268 people part-time	Same as current contributions

			The majority of these employees are from the local project region. Employees have gained experience conducting the baseline, distribution and monitoring surveys for the project which is recorded in the databases. So far in total 36 tablets have been provided to the local teams to carry out the field surveys. Of the full-time employees, 11 were given senior roles as mandal coordinators to take charge of the field team in each district of the first project instance region. Many more people have been indirectly employed through the project activity. These include: • 150 people employed in the country for manufacturing of the project ICS • 68 employed for transporting the ICS across the country Information on the stakeholder events conducted for the first instance are presented in Appendix 1, and include all local speakers invited to present to the communities. Employees were given opportunities here to help organise and host the events.	
 RESPONSIBLE CONSUMPTION AND PRODUCTION	12.2.2 Domestic material consumption, domestic material consumption per capita, and domestic material consumption per GDP	Implemented activities to decrease	Baseline studies state that • 100% of families source firewood for their stoves from local forest areas in the project region This has resulted in widespread deforestation as depicted in Figure 8. Given the scale at which deforestation is happening, owing to all villages in the region adopting similar stove usage practices, responsible consumption of firewood is critical.	Same as current contributions

			Based on the average firewood consumption used by traditional stoves reported in the baseline database of 18.54kg per day or equivalently 6.77 tonnes per year, and using the methodology (15% baseline efficiency) and technical specification of the Tramontana ICS detailed in section 1.12, we calculate across all ICS distributed (accounting for the individual days of operation), a total saving of 52,644 tonnes of firewood under the project (see distribution database).	
 CLIMATE ACTION	13.2.2 Total greenhouse gas emissions per year	Implemented activities to decrease	As presented in section 7.5, for the monitoring period the total emissions reductions from all operational ICS is 62,743 tCO₂e. All ICS distributed are currently being used by families on a regular basis. No additional monitoring was required for this SDG as the calculation is already a central requirement to the verification and generation of VCU's under Verra.	Same as current contributions
 LIFE ON LAND	15.1.1 Forest area as a proportion of total land area 15.2.1 Progress toward sustainable forest management 15.3.1 Proportion of land that is	Implemented activities to increase 15.1.1 and 15.2.1 Implemented activities to decrease 15.3.1	As highlighted for SDG12, baseline surveys confirm that all firewood sourced for use in stoves is collected from local forest areas and contributes to widespread deforestation in the region. It is observed that the average distance to collect firewood is 3.31km and is steadily increasing year-on-year. Starting from the monitoring output for SDG12, the total mass of firewood saved under the project over the first monitoring period was calculated as 52,644 tonnes. Assuming average dimensions of a	Same as current contributions

	degraded over total land area		forest tree in the local region of 14cm in diameter and height 4.57m (or 15ft), with spacings between trees of 3.5m, • a total of 1285 hectares of forest area has been preserved over the period through implementing the project (see calculations in distribution database).	
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1.19 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage in line with the methodology (see section 5).

Commercially Sensitive Information

All information for the purposes of project description and project implementation is included in this report which is made publicly available.

2. SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Pre-baseline surveys are conducted across an entire instance region to identify all eligible households. Subsequently multiple beneficiary stakeholder meetings are held to further engage with eligible ICS beneficiaries and convey more detailed information. Additional project stakeholders are directly employed or partnered with at the design and planning phase of a project instance. <u>Primary stakeholders:</u> local families receiving a Tramontana ICS, local employees such as field staff, local administrators (e.g. District Collectors), NGOs <u>Secondary stakeholders:</u> Outside of the local project areas, additional people are employed to support the project activities including manufacturing, transportation etc.
Legal or customary tenure/access rights	As part of the project activities, close engagement with the local communities is facilitated to identify any potential concerns and legal conflicts in relation to the project. By means of access to a local project representative or through a designated phone number, stakeholders can raise any concerns or questions they have throughout the project. To date, no conflicts or issues have been raised. The project does not interfere with any legal or customary rights to territories and resources held by project stakeholders.
Stakeholder diversity and changes over time	The ICS beneficiaries under the project live in rural areas and are predominantly from indigenous and socio-economically disadvantaged communities. The vast majority are represented by the Scheduled Tribe/Adivasi or Scheduled Caste/Dalit community categories discussed in section 1.12. The makeup of the indigenous tribes within a project instance is similar and all follow similar customs and practices. This is not expected to change over time.

	As part of the project, local staff from the project region are employed and trained to undertake the activities on-site and engage with the communities. All primary and secondary stakeholders involved in the project are located in India.
Expected changes in well-being	The project aims to have a positive impact on stakeholder wellbeing. The social and economic benefits of the project are outlined in detail in Section 1.18 and are briefly listed here: • Reduced drudgery and exposure to risks including wild animal attacks, sexual assault, injuries from tree cutting/wood chopping • Increased time for skill/economic development • Improved home conditions and quality of life for families • Enhanced safety and reduced burns/injuries for children • Development of IT skills for the local teams employed • Employment opportunities • More hours available for productive income generating activities and schooling
Location of stakeholders	ICS beneficiaries under the first project instance are located in Andhra Pradesh state in India. The precise location of each beneficiary is recorded during ICS installation and is presented in the distribution database. Outside of the local project areas, the people employed to support the project activities (e.g. manufacturing, transport) are all located within the project boundary of India.
Location of resources	Many of the project communities undertake agricultural work on land which has been provided to them (with ownership) by the government. This land is typically in close proximity to the beneficiary's home. The project does not interfere with such territories of the stakeholders.

2.1.2 Stakeholder Consultation and Ongoing Communication

The objective of stakeholder consultation meetings is to present the Tramontana ICS-01 project to the local communities and gather feedback. This mainly involves highlighting the project objectives,

the key benefits of using the ICS device, the stakeholders involved, as well as direct engagement and undertaking question/answer sessions with the communities.

The ICS project is designed to distribute 400,000 ICS across multiple states of India, as a grouped project with separate project instances. Multiple stakeholder consultations will be conducted for each project instance to ensure engagement with the communities in each area.



Figure 9. Newspaper advertisement for the first stakeholder meeting of first project instance held on 14th November 2022

Date of stakeholder consultation	The stakeholder consultations for the first instance comprising of 11 mandals across the Alluri Sitarama Raju (ASR) district of Andhra Pradesh mainly occurred between 14th November 2022 and 4th December 2022. On average two meetings were conducted per mandal which ensured a uniform and extensive consultation with community stakeholders. The schedule followed for the first project instance, which included a total of 2231 attendees, is outlined in Appendix 1. Minutes and attendance sheets were taken for all meetings. Figure 11 illustrates a newspaper article discussing one of the stakeholder meetings conducted. Meetings with a similar format and structure will be undertaken across all project instances in the future.
Stakeholder engagement process	Prior to stakeholder meetings, pre-baseline assessments are carried out to identify potential eligible households in villages. Once an area has been scoped out and the preliminary data has been collected, the project team can proceed with organising stakeholder meetings and baseline surveys. All potential beneficiaries of an ICS within a mandal are invited to attend in-person stakeholder meetings, which are advertised in advance through newspapers (example shown in Figure 9), text message, pamphlets and/or word of mouth. At each meeting, banners in the local language containing information on the project and the Tramontana ICS are displayed around the venue. Every attendee is provided with a physical brochure containing similar information (presented in Figure 10), and also offered refreshments during the meeting. Several special guests are invited to the meetings such as MPs, District Collectors and Community Presidents. These guests have a strong influence in the local area owing to their position of responsibility, knowledge of the local community and good understanding of the challenges faced by the people. All stakeholder meetings are conducted following local customs and traditions. The meetings are held in the local language, which for the first project instance in Andhra Pradesh is Telugu. In each meeting the Project Proponent and other local stakeholders are first introduced by a member of the team. Then a detailed presentation is given on the project, including the main objectives and a comprehensive (non-technical)

	overview of the ICS including highlighting the social (particularly for women), economic, and environmental benefits of adoption. Furthermore, an outline is given of the distribution/monitoring phases of the project and the underlying carbon credit-based financing. One other aspect of the presentation is the emphasis on the responsibilities of all households with regards to correct and continued ICS use, together with the support recipients will receive regarding repairs and replacements. Following the main presentation, each special guest in turn gives a speech providing their own insight into the project and impact on the local community. Finally, attendees are given the opportunity to ask questions or share any concerns directly with the project team through a microphone handed to them (depicted in Figure 12). The size of each stakeholder meeting is dependent on the mandal population and response rate. A full list of all attendees at each meeting is recorded, including their signatures, contact number and address. In addition, the minutes of every meeting including stakeholder questions and the responses are recorded and documented (in English). All information recorded from meetings including detailed minutes, attendance sheets, stakeholder questions and photos are made available.
Consultation outcome	The stakeholder meetings for the first project instance were all successfully conducted. The project gained a very positive reception and overwhelming support from the local communities across all 11 mandals. All invited special guests were keen advocates of the project activity, communicating their strong support and highlighting the importance of such an initiative for marginalised and underrepresented communities. Many reiterated the message that the Project Proponent is the first ever organisation to provide significant support to their local community and that the improved cooking technology will be life changing. Attendees were given the opportunity to ask questions or raise their concerns during (and after) all stakeholder meetings. Some examples of questions raised by attendees are detailed in Table 7. No negative comments were made by attendees

	with regards to the project activity, and thus no changes were required to the general project design or operation.
Ongoing communication	People from the local area are employed as field staff and project coordinators. This enables them to connect properly with the local communities and understand their needs during the project life. All concerns or questions that occur following a stakeholder meeting, can be directly reported to a local mandal representative or voiced through a dedicated phone number. These formats are suitable given many local people in the communities are illiterate and unable to express their feedback in written format. All points raised will be added by the team to a local grievance register. The information in the register will be reviewed on a regular basis by the Project Proponent, with all concerns addressed and incorporated into the project on a merit basis. At least annual visits are made by local project representatives to the villages in each area through the project life.
Stakeholder input	All questions raised and corresponding answers given by the project team during stakeholder meetings are recorded in the form of minutes. All minutes and attendance sheets for the first project instance are provided to the VVB and Verra. Some of the most commonly asked questions for the first project instance are given in Table 7. The project team also made themselves available after all meetings for any further discussion points to ensure all attendees were satisfied. No changes were required to be made to the project design as a result of the questions/discussions. Any key points raised from future stakeholder meetings will be recorded and if justified incorporated into the project design.

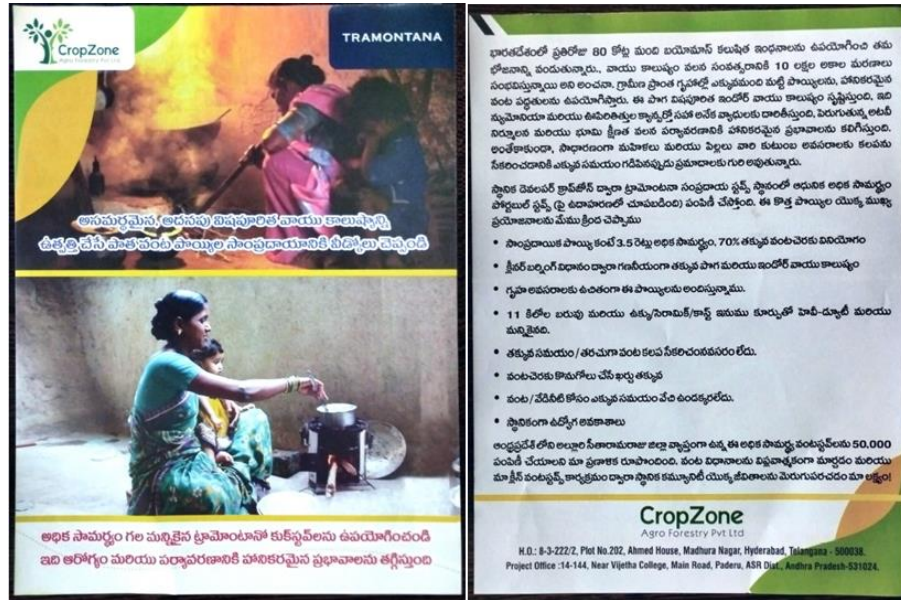


Figure 10. Brochure in the local language distributed at stakeholder meetings providing information on the project and ICS including its benefits



Figure 11. Local newspaper article discussing one of the stakeholder meetings conducted



Figure 12. Stakeholder meetings depicting a Tramontana presentation, special guest speech, high turnout, attendance sheet signing, community questions and ICS demonstration

Table 7. Some of the most common questions raised by local community members and responses given by the project team for various stakeholder meetings in the first project instance

Name of attendee	Question	Local coordinator response
Vanthala Lavanya from Baski village (Araku Valley)	How do traditional cookstoves and the Tramontana ICS differ?	As highlighted in the presentation, unlike traditional cookstoves which produce a lot of toxic localised air pollution and are low in quality, Tramontana ICS are made using steel/ceramic and fully combust the firewood to generate more heat and significantly less smoke. It also saves time on fuel gathering for the family as there is an estimated 60% reduction in firewood requirement. A list of the advantages can be seen in the brochure provided and on the banners displayed around the venue. You can discuss further details with any of the local representatives present here following this meeting.
G. Chandramma from Y.N Pakalu village (Koyyuru)	How do we fire up these ICS?	The Tramontana ICS can be fired up in the same way as your current cookstoves. The advantage is that you do not need to drastically change the way you cook or use the stove. We will provide instructions verbally and in a brochure format with every ICS during distribution, as well as give physical demonstrations, to help you safely and effectively operate and maintain the ICS. Also, a contact number will be provided in case you are having any issues or require advice.
B. Kondamma from Pedalochali	How much weight can be put on the Tramontana ICS for cooking?	We estimate you can put up to 15 kg of weight on the Tramontana ICS, which should be more than sufficient to support cooking (and other activities) for the entire family.

village (G. Madugula)		
D. Saraswathi from Dokuluru village (Paderu)	If there are 10 members in a family, how will a single ICS be sufficient?	A single ICS may in fact be sufficient due to its increased thermal efficiency and higher cooking capacity. However, we will consider providing an additional stove if the amount of firewood you are currently using is significantly higher than the average family in the community. For this we would have to take further fuel weight measurements and other details for evaluation ¹² .
Konudi Pushpalatha from Dabavalasa village (Hukumpeta)	How long are the Tramontana ICS under warranty?	The average lifespan of the Tramontana ICS is estimated to be 7 years. We will provide you with a contact number at the time of distribution which you can contact if at any time you are experiencing problems with your ICS. We will provide maintenance, repair and replacement services due to any malfunction or damage to the ICS free of cost. Our team will also conduct at least annual household visits to ensure you are using the ICS and are satisfied with it.

2.1.3 Free Prior and Informed Consent

Obtaining consent	During stakeholder meetings, the benefits of the ICS devices are described and safe usage of the ICS is demonstrated. The meetings also represent an opportunity for stakeholders to raise any concerns or ask questions, as discussed in Section 2.1.2. These measures ensure that stakeholders are well informed, able to relay their views, and project activities are fully transparent. During distribution of the ICS, an agreement is signed between the Project Proponent and each beneficiary. The terms of the
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¹² See section 5.4 for details on accounting for multiple devices per household in emissions reduction calculations

	agreement are outlined at a high level during stakeholder meetings, and in a more detailed manner during ICS distribution.
Outcome of FPIC	All beneficiaries provided consent to their involvement in the project and the terms of the license agreement. The feedback was overwhelmingly positive for project participation based on the stakeholder meetings and pre-baseline surveys conducted. The project has not encroached on land, relocated people without consent, or forced physical or economic displacement. No conflicts have been recorded under the project to date.

2.1.4 Grievance Redress Procedure

Development process	The grievance redress procedure was developed based on consultations with stakeholders during the planning stages of the project. The process ensures that any grievances raised by stakeholders are respected, recorded and addressed.
Grievance redress procedure	Any concerns or questions stakeholders may have following stakeholder meetings and during the project life, can be directly reported to the local mandal representative or communicated through a dedicated phone number. All points raised will be added by the team to a local grievance register. The information in the register will be reviewed on a regular basis by the Project Proponent, with all concerns addressed and incorporated into the project subject to merit.

Grievances received	Resolution and outcome
None	No grievances were raised during the monitoring period. The local grievance register is accessible and checked on a regular basis by the Project Proponent.

2.1.5 Public Comments

Comments received	Actions taken
None	No public comments were received during the public comment period which was from 16-June-2023 to 17-July-2023

2.2 Risks to Stakeholders and the Environment

Potential Risks	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	Risk identified	Multiple preventative measures have been taken to minimise risks associated with stakeholder participation. During stakeholder meetings, the benefits of using an ICS device and following sustainable practices are clearly communicated to families, alongside outlining the planned project activities. This ensures that the community members understand the goals of the project and are educated on the benefits to them generated by it. Throughout the project, regular monitoring ensures that the project status is continuously assessed, including tracking the number of ICS that are operational following installation. This ensures that any ICS found not to be in use is quickly identified and removed from emissions reduction calculations, and direct communication is made with the corresponding beneficiary to understand the reasons for not using their ICS. Continuous monitoring and the employment of local project staff also ensure that any challenges in regards to the operation of an ICS (such as breakdown of the ICS, severe damage etc) are swiftly resolved.
Working conditions	Risk reduced from baseline	The Tramontana ICS project does not attempt to change the working customs of local communities, including the general stove type and method of use. This ensures confidence and comfort to households when using an ICS. The ICS device is associated with cleaner combustion characteristics and increased thermal efficiency. This improves working conditions in the home, as the amount of indoor air pollution is significantly reduced, and more

		family time is available owing to time savings when collecting firewood. It also enhances safety and reduces burns/injuries due to the compact nature and improved design of the ICS.
Safety of women and girls	Risk reduced from baseline	The project directly improves the safety and well-being of women and girls who are the ones mostly undertaking stove-related tasks, including usage, firewood collection and cleaning. The project reduces drudgery and exposure to risks including wild animal attacks, sexual assault, injuries from tree cutting/wood chopping etc. that are prevalent in the project area. These issues were widely reported in baseline surveys for the first project instance (see section 3.4). The benefits to female members observed during monitoring are highlighted under SDG 5 (Gender Equality) in Table 6.
Safety of minority and marginalized groups, including children	Risk reduced from baseline	During stakeholder meetings, explanations and demonstrations of the ICS are given to ensure that families are clear on its correct, safe operation and any risk of injury is minimised. Any clarity needed or difficulties faced during operation can also be discussed with a local representative or using the telephone number provided. The ICS device has generally been designed to be compact and safe to ensure that there is minimal risk of burn or injury to users.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	Risk reduced from baseline	The Tramontana ICS significantly reduces toxic indoor air pollution through its improved design and cleaner combustion characteristics. This has immediate benefits for the project families, including reducing respiratory and retinal diseases. More details of the benefits observed during monitoring can be found in Table 6 which records the SDG benefits observed.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	As part of the project activities, a training program is implemented for all field staff and coordinators employed. The training enforces strict rules regarding harassment and discrimination. Should any accusations be made by beneficiary families or other villagers, the specific employed individuals will be evaluated and immediately removed from the programme if found to be at fault. Since the onset of the project, no project stakeholders have reported any form of discrimination or sexual harassment.
Management experience	The Project Proponent has extensive expertise in risk management in financial markets, project development and the management of large teams. The local partner Cropzone, has strong relationships with the local communities and some experience in on-the-ground logistics and project implementation. The Project Proponent is the coordinator and manager of the entire project. It makes all major decisions required in the project, and provides continuous guidance and support to the local teams. The Project Proponent provides extensive training and support to all project teams, which is discussed in detail in section 6. The more senior mandal coordinators employed are given specific management training for managing the local field teams and tackling any issues when engaging with beneficiaries.
Gender equity in labour and work	All employees under the project have been paid fairly and in accordance with local customs and workers' rights. Employment of local community members and improving their livelihood and skillset is a central component of the project, particularly given the lack

	of opportunities available to the marginalised project communities. Where possible the project looks to employ female team members to improve the gender representation that are in work, which is typically low in the project region. The reduced time needed to collect firewood as a result of the project activity has resulted in more opportunities for women to spend time in contributing to the main family income source, undertaking community work and their children to spend in education.
Human trafficking, forced labour, and child labour	Human trafficking, forced labour and child labour are strictly forbidden in the project. Background checks are conducted on all staff. A strict 18+ age policy is also followed. All work policies and procedures including in relation to human trafficking, forced labour and child labour are communicated to all teams involved in the project work during training.

2.3.2 Human Rights

The project recognises, respects, and promotes the protection of the rights of all local communities and employees in line with applicable international human rights laws, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples.

2.3.3 Indigenous Peoples and Cultural Heritage

All project activities are conducted following local customs/traditions, respecting the area's cultural heritage. All stakeholder meetings and surveys are also conducted in the local language and using team members employed from the local community.

2.3.4 Property Rights

Rights to territories and resources	The project does not engage with territory, resource or property rights. The ICS are provided free of cost with a licensing agreement, as outlined in Section 2.1.3.
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	As part of the user license agreement, the beneficiaries give up and transfer the rights to claim carbon credits to the Project Proponent in return for usage of the ICS.
Respect for property rights	The project fully respects all property rights and does not impact them through the project activities carried out

2.3.5 Benefit Sharing

This section is not applicable as the project does not impact property rights as discussed in Section 2.3.4.

2.4 Ecosystem Health

No negative environmental impacts were identified for the project. An Environmental Impact Assessment (EIA) is not required for the project as there are only positive environmental outcomes.

Impact	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	Through the improved design and increased thermal efficiency of the ICS, there is a reduced firewood requirement. This results in reduced deforestation/forest degradation and a positive impact on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The reduced deforestation/forest degradation resulting from the lower firewood requirement, leads to less soil degradation and erosion.
Water consumption and stress	No risk identified	The better quality soil conditions as a result of the project activities lead to improved soil water retention. Less water is needed for cleaning utensils and maintenance of the household owing to the cleaner combustion of the ICS and

		minimised smoke/soot generation. These points were also highlighted in the monitoring database by beneficiaries.
Usage of fertilizers	No risk identified	The project activities do not have a measurable impact on the usage of fertilizers.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

N/A

2.4.3 Ecosystem conversion

N/A

3. APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type	Reference ID	Title	Version
Methodology	VMR0006	VMR0006 Energy efficiency and fuel switch measures in thermal applications ¹³	1.2
Methodology	CDM AMS-II G	Energy efficiency measures in thermal applications of non-renewable biomass ¹⁴	13.1
CDM Tool	CDM TOOL30	Calculation of the fraction of non-renewable biomass	4.0
CDM Tool	CDM TOOL33	Default values for common parameters	2.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VMR0006	Project activities must introduce energy efficiency and fuel switch measures in thermal applications (including cookstoves, ovens and dryers) that: - increase thermal efficiency to reduce the consumption of non-renewable biomass, or	The project activity involves the distribution and installation of thermally efficient ICS in rural households in India which use non-renewable biomass

¹³ https://verra.org/wp-content/uploads/2023/07/VMR0006-v1.2_PDF.pdf

¹⁴ <https://cdm.unfccc.int/UserManagement/FileStorage/6TOUCX21D0BHNVRZFWMEKALY94GS7>

	- switch from fossil fuel (coal or kerosene) to renewable biomass in new or existing improved thermal energy generation units.	
VMR0006	The project activities must be implemented in households, community-based kitchens, institutions (e.g. schools) or small and medium sized enterprises (SMEs)	The project activity is implemented in households
CDM AMS-II G	The project cookstove must have a specified rated thermal efficiency of at least 25% as per the manufacturer's specifications, be single-pot or multi-pot and portable or in-situ	The Tramontana ICS installed under the project activity is a high-quality single-pot portable cookstove, which uses woody biomass as a fuel input in line with manufacturer specifications. The ICS has an efficiency of 36.01% obtained through independent verification via laboratory testing conducted at the Department of Mechanical Engineering at Jorhat Engineering College (see certificate in Appendix 2). Any new ICS models which may be included in future instances will also meet the requirements of this methodology criteria.
VMR0006	Project activities characterised as Projects and Large Projects can apply this methodology	The project activity involves the distribution of 400,000 ICS across India and the estimated annual average GHG emissions reductions generated from this is 1,310,160 tCO ₂ e (see section 5). As a result, the project meets the condition of Large project under the standard (which corresponds to

		over 300,000 tCO ₂ e per annum) and therefore can apply this methodology.
CDM AMS-II G	The aggregate energy savings of a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input	The energy savings for a single project activity is found to be below 60 GWh per year. The calculations meeting such condition are given in the Distribution spreadsheet provided.
CDM AMS-II G	Non-renewable biomass has been used in the project region since 31 st December 1989, using survey methods or referring to published literature, official reports or statistics	According to the UN Food and Agriculture Organisation, there was significant demand for firewood in India observed over the period 1970 to 1995 (Table 8 ¹⁵), and also a large percentage share of the energy consumption in rural households coming from firewood (Table 9 ¹⁶). The high demand has been a key driver of forests being cleared/degraded, as presented in the State of Forestry reports from Forest Survey of India ¹⁷ . Over the period 1993 to 1997, 671,000 hectares or equivalently over 1% of total forest cover (dense and open forests) was lost in India. In addition, 1,831,000 hectares of dense forests was either degraded or destroyed. In the first project instance location of Andhra Pradesh, the total forest cover loss over the same period was estimated to be 396,600 hectares.

¹⁵ <https://www.fao.org/forestry/statistics/80570/en>

¹⁶ <https://wgbis.ces.iisc.ac.in/energy/HC270799/RWEDP/acrobat/fd49.pdf>

¹⁷ <https://fsi.nic.in/index.php>

		Therefore it is clear that biomass from non-renewable sources has been in use in the project region since at least as far back as 31st December 1989. For the period beyond 1997 data sources such as the Global Forest Watch¹⁸ show that use of non-renewable biomass has increased. The Global Forest Watch report between 2001 to 2023 states that India lost 2.33Mha of tree cover, equivalent to a 6% decrease in tree cover since 2000.
CDM AMS-II G	For cases where the biomass is sourced from renewable sources, the project participants should use a corresponding Type I methodology	Not applicable as biomass is not sourced from renewable sources under the project activity
VMR0006	For fuel switch activities, the project must exclusively use renewable biomass and apply the detailed conditions stated in the methodology	This condition is not applicable as the project activity does not involve fuel switching. Instead, the activity involves reducing the consumption of non-renewable biomass by replacing traditional cookstoves with high quality thermally efficient ICS.
CDM AMS-II G	The VCS project document shall explain the proposed method for the distribution of project devices including the method to avoid double counting of emission	Each ICS installed under the project shall be identifiable by the Project Proponent logo and a unique serial number which is displayed on the ICS itself as a QR code and in written form.

¹⁸ <https://www.globalforestwatch.org/dashboards/country/IND/?category=forest-change&location=Wyj3VudHJ5liwiSU5EI0%3D>

	reductions such as unique identifications of product and end-user locations (e.g. programme logo)	This is linked to individual users using the Tramontana ICS software application which scans the QR code and records corresponding beneficiary information including personal details, address, date of installation, geolocation of household, user/ID photos etc (see section 1.12.1 for full list). The data will be verifiable by the VVB through an online database of distribution records. These project design measures eliminate the risk of double counting. Further details on the ICS distribution process can be found in section 1.12.
CDM AMS-II G	The VCS project document shall also explain how the proposed procedures prevent double counting of emission reductions, for example, to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices	A standardised license agreement is signed physically (with written signature or thumbprint) by each ICS beneficiary which includes (amongst other conditions) provisions for ownership of all emissions reductions generated by the ICS to be solely with the Project Proponent. The license agreement is generated through the Tramontana ICS software application on a tablet which is then signed by the end user (see Figure 2). A signed pdf copy of all agreements is available in the distribution database. No other stakeholders or parties involved in the project have any rights to the carbon

		credits generated from the GHG emissions reductions.
CDM TOOL30	This tool may be used by: a) DNAs to submit region- or country-specific default fNRB values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or b) project participants to calculate project- or PoA-specific fNRB values.	The tool is used by the Project Proponent to calculate project specific fNRB values, as presented in section 5. Thus applicability condition b) is met.
CDM TOOL33	This tool shall be applied in conjunction with the methodologies which refer to this tool to source the default values to estimate the baseline emissions (e.g. from the use of diesel for off-grid power generation, from the use of kerosene for lighting, from the use of woody biomass for cooking).	The CDM AMS-II.G v13.1 methodology which specifically refers to CDM Tool 33 is used to obtain the pre-project device efficiency value: “project participants may also conservatively assume that the efficiency of pre-project device is the highest among the default efficiency values provided in TOOL33”. Given the use of three-stone fires or cookstoves with no improved combustion air supply (i.e. without a grate or chimney) is used in the project baseline, a value of 0.15 is applied. As such, the applicability condition is met.
CDM TOOL33	The default values as contained in section 5 of this	The project start date is prior to 10 th March 2025 and therefore

	tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the default values after this date, including in cases where further default values are added to this tool through revisions of this tool before this date	the default value of 0.15 used from CDM TOOL33 for the efficiency of pre-project cooking devices is valid. Any project activity occurring after 10th March 2025, will use the latest default value available.
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Table 8. Total firewood demand in India over the period 1970 to 1995¹⁵

Year	1970	1975	1980	1985	1990	1995
Firewood Demand (million m ³)	154.99	173.36	194.42	214.49	237.63	260.77

Table 9. Estimate share of energy consumption of firewood in rural households¹⁶

Type	% share in 1978 -79	% share in 1992-93
Firewood logs	18.95	32.49
Firewood twigs	35.62	29.11

3.3 Project Boundary

The boundary for the grouped project can be represented by the geographical boundary of India as presented in Figure 13 and also outlined in section 1.13.

Each project instance will also have its own separate defined boundary within the overall boundary of the grouped project. For the first project instance this corresponds to the Paderu division of the Alluri Sitarama Raju district of Andhra Pradesh depicted in Figure 7 and represented by approximate coordinates of latitude 18.011959 and longitude 82.555030.



Figure 13. Delineation of geographical boundary (red) within which all activity under the grouped project is contained

The greenhouse gases to be included in the project boundary are presented in the table below and the corresponding impact of introducing the Tramontana ICS is illustrated in Figure 14.

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from the use of non-renewable biomass	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Major source of emissions
		N ₂ O	Yes	Major source of emissions
		Other	No	No other sources identified

	Source	Gas	Included?	Justification/Explanation
Project	Emissions from the use of non-renewable biomass	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Major source of emissions
		N ₂ O	Yes	Major source of emissions
		Other	No	No other sources identified

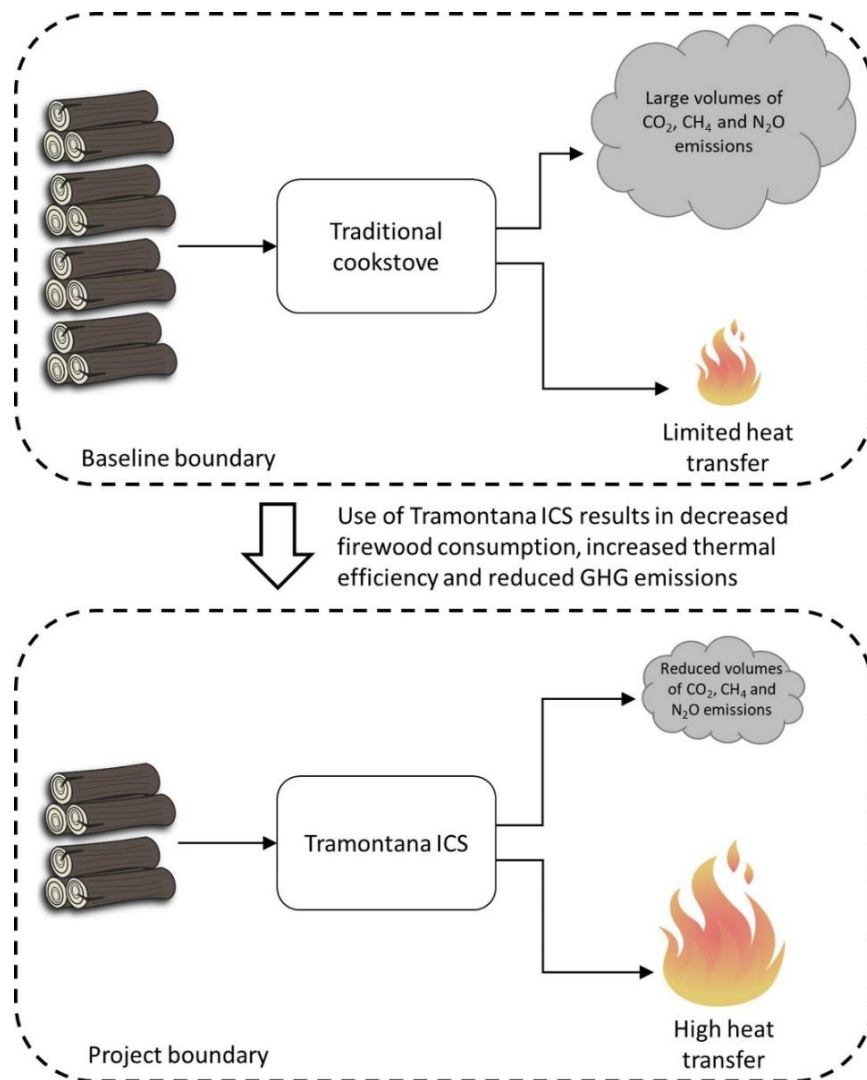


Figure 14. Depiction of the impact of introducing the Tramontana ICS on GHG emissions, firewood consumption and thermal efficiency in the project region

3.4 Baseline Scenario

The baseline or pre-project scenario is the continued use of woody biomass by local communities to meet their thermal energy needs such as for cooking, heating and boiling water. Use of rudimentary traditional cookstoves is prevalent, most commonly in the form of fixed mud clay stoves (Figure 1 and Figure 15) or three-stone stoves (Figure 3). Usage of these stoves has generally been commonplace in the project areas for generations. Some broader information on the baseline scenario is provided in sections 1.12 and 1.14.

Despite all project activity instances sharing some comparable characteristics, separate baseline surveys are to be conducted for each instance in order to gauge an accurate representation of the current practices in that specific region. A similar approach to the baseline surveys conducted for the first project instance outlined below will be followed for other instances.

For the first project instance, baselines surveys were conducted across 11 mandals in the Alluri Sitarama Raju district of Andhra Pradesh. This occurred between 3rd November 2022 and 12th December 2022. Pre-baseline surveys were initially conducted to estimate the approximate number of potential ICS beneficiaries in each village. Applying random sampling (minimum sample size 100) to the pre-baseline data to identify villages to be included in the baseline, and with significant oversampling to maximise reliability, approximately 3% of all potential beneficiaries for the instance were identified and interviewed as part of baseline surveys. This amounted to in total 1793 baseline surveys being conducted over the period.

To understand the baseline situation across the project instance area, detailed surveys were conducted which recorded information including: respondent family details (which includes the role of each member in stove usage/fuel sourcing), current stove usage practices, fuel collection and consumption patterns (with measurements), the current issues faced and the perception of adopting a new ICS. All surveys were carried out by trained local field teams employed for the project, using tablets containing the Tramontana ICS software application. All baseline records were signed, submitted and successfully synced to a Google cloud database before being automatically translated into an Excel sheet. The sheet was then checked by an office team to identify any incorrect entries and immediately have them resurveyed. The final baseline sheet is submitted to the VVB and Verra for the purposes of validation.

From the baseline survey results for the first project instance, it is found that the vast majority of families (79%) are using traditional fixed mud clay stoves and a smaller proportion (20%) use three-stone stoves. Over 85% of survey respondents are using their stove inside the household. We have measured that 18.54kg per day (or equivalently 6.77 tonnes per annum) of firewood is being consumed per stove on average across all seasons, with all respondents confirming that wood as their only fuel source. In all cases firewood is sourced directly from forests. Every user carries their firewood by headload and the average distance from the source location was found to be 3.31km. In general, the distance to the source location has been increasing year-on-year as forest areas closer to households have been deforested.

The traditional stoves are used an average of 4.5 hours per day all year round for the purposes of cooking, boiling water and in many cases (83%) also room heating. Note that in 10% of the cases individuals also used their stoves to prepare food/beverages for sale. In addition to the above, an

estimated 12 – 15 hours per week is spent sourcing, carrying and processing (e.g. cutting, shredding) the firewood, which highlights the overall time-consuming nature of stove related tasks. It is exclusively women who were using the stoves (aside from a few cases where no women lived in the household), with the firewood gathering also undertaken by them, sometimes with the help of other family members. Generally, it was clear that stove-related activities formed the primary role of female members in the community.

Over 98% of respondents stated that they were not happy with their current stove and accordingly very few stated that it was not a priority to change. All participants confirmed that they did not own an ICS and would be interested to receive a new ICS if offered one.

A number of key issues were widely reported across the baseline surveys. These included: regularly experiencing wild animal attacks in the forest (including snakes, bears, boars and dangerous insects), injuries when cutting wood using tools/axes, sexual assault, allergies from exotics plant species, toxic localised smoke causing retinal and breathing issues, and physical pain in undertaking the physically demanding tasks of cutting trees, carrying firewood long distances in hilly areas and sitting in a fixed position adjacent to the stove for prolonged periods.

Almost all respondents (91%) stated that there is a lot of localised smoke produced by their traditional stoves, that the associated tasks are time consuming giving little time for other activities, and that a large quantity of firewood is required to be regularly collected/used. Despite all respondents knowing that kitchen smoke has a negative impact on health, they have little ability to control it. When questioned on why they use traditional stoves which have such issues, the main reasons given were that they are not aware of any other stoves, alternative fuel sources are too expensive, firewood is abundantly available and/or that their current stove has been at least manageable in terms of handling/maintenance.

All baseline respondents confirmed having a Below Poverty Line/ration card, and over 99% confirmed availability in physical form at the time of surveying. Where available these cards will be photographed at the time of ICS distribution. Finally, over 99% of the respondents stated that their family's main work is in agriculture and the total income per family was found to average out to under a US dollar a day.



Figure 15. Baseline scenario depicting a traditional mud clay stove, firewood being transported by headload, cooking using a traditional stove and Tramontana meeting with ICS beneficiaries from a local indigenous community

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

3.5.2 Additionality Methods

The methodology uses the Activity Method to demonstrate additionality. This involves:

Step 1: Regulatory Surplus

The project is designed and implemented on a voluntary basis by the Project Proponent and therefore it is a voluntarily coordinated action.

There is no mandated government program or policy in the host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is also not mandated by any law, statute, or other regulatory frameworks, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework.

Step 2: Positive List

The positive list represents the applicability conditions (outlined below) from the methodology which must be satisfied for a project to be deemed additional.

Condition 1: The project activity installs or distributes the project devices at zero cost to the end user and has no other source of revenue other than carbon credits.

Condition 2: The project activity is not implemented as part of a government scheme or supported by multilateral funds.

The project ICS (and any other ICS that may be included in future instances) will be distributed and licenced to beneficiaries free of charge under the project activity, with no sources of revenue generated from the project activity other than from the sale of GHG carbon credits. The project activity is independently undertaken by the Project Proponent on a voluntary basis and is not affiliated with any government schemes or supported by multilateral funds. As the project activity satisfies applicability conditions 1 and 2 above, it can be deemed as being additional.

Step 3: Project Method

Not applicable as the ICS are provided free of charge to all end users

3.6 Methodology Deviations

The project activity has not taken any deviations from the applied methodology VMR0006.

4. IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

Up to the end of the monitoring period of 23rd August 2023, a total of 46,160 ICS were distributed as part of the first project activity instance (PAI) to households in the ASR district of Andhra Pradesh, India. Note that each instance will constitute a new geographical region, but not necessarily a new state of India. Each household in the first instance received one portable ICS of the type illustrated in Figure 4, and all beneficiaries confirmed using their ICS for their daily activities for all days of the week during the period, as detailed in section 7 on monitoring. All baseline stoves had been removed or demolished, and there were no other form of baseline stoves observed to be in use. Therefore, the new ICS provided was found to meet the equivalent thermal energy needs of the pre-project stove. Two of the project beneficiaries are presented in Figure 16 together with their Tramontana ICS.

No significant events occurred during the monitoring period which affected the GHG emissions reductions or required a change to the monitoring framework presented. There has also been no change to the entities involved in the project including the Project Proponent.



Figure 16: Beneficiaries using the Tramontana ICS for their daily activities

5. QUANTIFICATION OF ESTIMATED GHG EMISSIONS REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net emissions reductions achieved by the project. Please refer to section 5.4.

5.2 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emissions reductions achieved by the project. Please refer to section 5.4.

5.3 Leakage Emissions

Leakage will be estimated using a default factor 0.95 (see section 5.4) in accordance with the AMS-II G methodology. This value is considered conservative, as there is little incentive for non-project users to switch fuels or increase consumption given that woody biomass is already generally widely available and zero cost across the project region.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

ICS are introduced as an energy efficiency measure in the project, therefore Equation 1 of the methodology VMR0006 is applied to calculate the net GHG emissions reductions.

$$ER_y = \sum_i \sum_j B_{y,sav,i,j} \times N_{0,i,j} \times n_{y,i,j} \times \mu_y \times f_{NRB} \times NCV_{wood_fuel} \times (EF_{wf,CO2} + EF_{wf,nonCO2}) \times Adj_{LE} \times (1 - u_d) \quad \text{Equation 1}$$

where

i	=	Indices for the situation where more than one type/model of ICS is introduced to replace baseline stoves
j	=	Indices for the situation where there is more than one instance of ICS of type i
ER_y	=	Emissions reductions during year y (tCO ₂ e)
$B_{y,sav,i,j}$	=	Quantity of woody biomass saved by an ICS of type i in instance j during year y (tonnes)
$N_{0,i,j}$	=	Number of ICS devices of type i and batch j commissioned (number)

$n_{y,i,j}$	=	Proportion of commissioned ICS devices of type i in instance j that remain operating in year y (fraction)
μ_y	=	Adjustment to account for any continued use of pre-project devices during year y (fraction)
f_{NRB}	=	Fraction of woody biomass that can be established as non-renewable biomass (fraction)
NCV_{wood_fuel}	=	Net calorific value of the non-renewable woody biomass that is reduced (TJ/tonne)
EF_{wf,CO_2}	=	CO ₂ emissions factor for the use of non-renewable woody biomass in the baseline scenario (tCO ₂ e/TJ)
$EF_{wf,nonCO_2}$	=	Non-CO ₂ emissions factor for the use of non-renewable woody biomass in the baseline scenario (tCO ₂ e/TJ)
Adj_{LE}	=	Adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity (fraction)
u_d	=	Uncertainty deduction for f_{NRB} (fraction)

Following the AMS-II G methodology, the quantity of woody biomass saved due to the installation of ICS is estimated using

$$B_{y,sav,i,j} = B_{old,i,j} \times \left(1 - \frac{\eta_{old,i,j}}{\eta_{new,i,j}} \right) \quad \text{Equation 2}$$

where

$\eta_{old,i,j}$	=	Efficiency of the baseline cookstove (%)
$\eta_{new,i,j}$	=	Efficiency of ICS of type i in instance j for year y determined through conducting water boiling tests (WBT) through the project life (%)
$B_{old,i,j}$	=	Annual quantity of woody biomass which would have been used by the baseline cookstove in the absence of the project activity in tonnes

Though unlikely, if at any time a household requests to receive more than one ICS owing to, for example a very large family size, the Project Proponent will consider the request and evaluate upon collecting further details. This would include additional information on ICS/fuel usage patterns, including undertaking fuel weight measurements. In the case of deciding to distribute more than one ICS to a household, the following calculation from the methodology is applied to determine the baseline fuel consumption per device.

$$B_{old} = \frac{B_{old,H}}{n_{HH}} \quad \text{Equation 3}$$

where

$B_{old,H}$ = Annual quantity of woody biomass which would have been used by the household in the absence of the project activity in tonnes

n_{HH} = Number of ICS distributed to the household

All of the following calculations for f_{NRB} are provided in an Excel sheet separately.

Following CDM TOOL30, the fraction of non-renewable biomass is calculated using the following equation:

$$f_{NRB} = \frac{NRB}{NRB + RB} \quad \text{Equation 4}$$

where

f_{NRB} = Fraction of non-renewable biomass in the applicable area in the relevant period (fraction or %)

NRB = Quantity of non-renewable biomass consumed in the applicable area in the relevant period (tonnes)

RB = Quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)

The quantity of non-renewable biomass (NRB) is determined as:

$$NRB = H - RB \quad \text{Equation 5}$$

where

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

The total consumption of woody biomass (H) is calculated using the following equation, accounting for all consumption within the applicable area (which includes timber and industrial consumption):

$$H = HW \times N + CE + NE \quad \text{Equation 6}$$

where

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes/household)

N = Number of households consuming wood fuel within the applicable area in the relevant period (number)

CE = Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE = Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

Parameter	Value	Source	Reference
HW	6.77 tonnes/hh/year	Baseline Survey – Instance 1	Average pre-project fuel usage
N	9,193,791	2011 Census of India ¹⁹	See f_{NRB} Excel sheet for exact calculation
CE	0	Data not available, so conservatively assuming zero	-
NE	78,345 tonnes	Calculated from India State of Forest Report 2019 ²⁰	See f_{NRB} Excel sheet for exact calculation

$$H = 6.77 \times 9,193,791 + 0 + 78,345$$

$$H = 62,308,098$$

The quantity of renewable biomass (RB) is determined as:

$$RB = \sum_i (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum_i (MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

Equation 7

where

$MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of forest areas in the relevant period (tonnes/ha/yr)

$MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of other land areas in the relevant period (tonnes/ha/yr)

$F_{forest,i}$ = Extent of forest in sub-category i in the relevant period (ha)

$F_{other,i}$ = Extent of other land in sub-category i in the relevant period (ha)

$P_{forest,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in subcategory i) in the relevant period (ha)

¹⁹ <https://new.census.gov.in/census.website/data/HH2011>

²⁰ <https://ruralindiaonline.org/en/library/resource/india-state-of-forest-report-2019-volume-ii/>

$P_{other,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category i) in the relevant period (ha)

Parameter	Value	Source	Reference
$MAI_{forest,i}$	4.63 tonnes /ha/year	2006 IPCC Guidelines for National Greenhouse Gas Inventories ²¹	See f _{NRB} Excel sheet for exact calculation
$MAI_{other,i}$	4.63 tonnes /ha/year	2006 IPCC Guidelines for National Greenhouse Gas Inventories ²¹	See f _{NRB} Excel sheet for exact calculation
$F_{forest,i}$	3,725,800 ha	India State of Forest Report 2019 ²⁰	p3 “Recorded Forest Area”
$F_{other,i}$	893,200 ha	India State of Forest Report 2019 ²⁰	Table 11.1.13 (Extent of TOF)
$P_{forest,i}$	3,195,900 ha	India State of Forest Report 2019 ²⁰	p3 “Reserved Forest”
$P_{other,i}$	0	Data not available, so conservatively assuming zero	-

$$RB = (4.63 \times (3,725,800 - 3,195,900)) + (4.63 \times (893,200 - 0))$$

$$RB = 6,581,838$$

Therefore NRB can be calculated:

$$NRB = 62,308,098 - 6,581,838$$

$$NRB = 55,726,261$$

And therefore, f_{NRB} is calculated to be:

$$f_{NRB} = \frac{55,726,261}{55,726,261 + 6,581,838}$$

$$f_{NRB} = 0.894$$

²¹ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

The following assumptions are applicable for the project activity to help estimate ex-ante emissions reductions:

- Project activity installs up to 400,000 ICS across the project region
- Commencement of ICS installation for a project activity instance is considered as the operational date of that project instance, and the date of installation of the last ICS within an instance determines the expected operational lifetime of the instance
- An expected average lifespan for an ICS of 7 years. This can be extended with regular maintenance, and information on any lifetime extensions will be reported during monitoring.
- An annual stove loss rate of $L = 0$, as the project aims to provide maintenance of all ICS and repair/replace in case of any damage or stove failure. Any issues found by an ICS user can be communicated in person or through a dedicated contact number provided. A physical check is also undertaken with every household at least once a year. Note that the parameter L exists for ex-ante calculation purposes only and is not directly determined for the project. Instead it is incorporated within the parameter $N_{y,i,j}$ which is monitored through the project life.
- B_{old} calculated based on averaging the baseline surveys conducted prior to ICS installation for each instance. Here it is estimated as 6.77 tonnes per year which corresponds to the output for the first project activity instance (see section 3.4).
- A value of $\mu = 1$ as all pre-project stoves are aimed to be removed or demolished when beneficiaries receive a new ICS. The value of this parameter for the project will however be based on actual monitoring survey records where on-site checks are undertaken for use of pre-project (or other) stoves.
- A ramp-up schedule defined for ICS distribution with 100,000 over the first year followed by 200,000 per annum for subsequent years, reaching a total of 400,000 ICS
- A single ICS model distributed across the entire project area and therefore i fixed. Note that if at a later date, an additional ICS model is included or an improvement is made to the existing ICS, it will be detailed as part of the monitoring report and correspondingly accounted for in emissions reduction calculations.
- For simplicity, ex-ante calculation workings in this section do not delineate project instances j

To determine the proportion of commissioned devices of type i in instance j that remain operating in year y for ex-ante calculation purposes, we use

$$N_{y,i,j} = N_{0,i,j} (1 - yL) \quad \text{Equation 8}$$

where

- | | | |
|-------------|---|---|
| $N_{y,i,j}$ | = | Total number of ICS of type i in instance j operating in year y |
| L | = | Annual loss rate of stoves in the project area |

y_l = Years of operation

It follows that

$$n_{y,i,j} = \frac{N_{y,i,j}}{N_{0,i,j}} \quad \text{Equation 9}$$

Applying Equations 8 and 9 with the distribution schedule stated above with single stove type, for example for $y = 2$,

$$N_{2,1} = 100,000 \times (1 - 2 \times 0) + 200,000 \times (1 - 1 \times 0) = 300,000,$$

and assuming all ICS are utilised by beneficiaries,

$$n_{2,1} = \frac{300,000}{300,000} = 1$$

Therefore, the expected number of ICS operating each year is

y	$N_{y,1}$	$n_{y,1}$
1	100,000	1
2	300,000	1
3	400,000	1
4	400,000	1
5	400,000	1
6	400,000	1
7	400,000	1

The stove efficiency $\eta_{new,y,i}$ is determined through regular water boiling tests (WBTs) conducted through the project life, as discussed in section 6. However, for the purposes of ex-ante calculations, we apply a 1.57% linear annual reduction factor to estimate the efficiency of the project ICS in each year y as

y	$\eta_{new,1}$
1	35.22%
2	33.65%
3	32.08%
4	30.51%
5	28.93%
6	27.36%
7	25.79%

The methodology outlined in paragraph 37 (a) of AMS-II G is applied to determine the linear efficiency reduction factor. This involves taking the difference between the initial and final efficiencies, and dividing by the lifespan of the project device. Using the initial efficiency of 36.01%, a terminal efficiency of 25%, following AMS-II G guidelines, and a life span of 7 years, an annual efficiency reduction of 1.57% is calculated. Using this reduction factor, the efficiencies at the start and end of each year have been calculated. Each efficiency in the above table then represents an average of the efficiencies at the start and end of each year. The full calculation of this is provided in the Ex-Ante calculation sheet.

Based on the annual ICS efficiency estimates above, Equation 2 is applied to determine the mass of woody biomass that is saved by an ICS in tonnes each year.

Using a value of $B_{old} = 6.77$ tonnes determined from baseline surveys and $\eta_{old} = 0.15$ in line with the methodology, and using the value for $\eta_{new,1}$ for $y = 1$ as an example from the table above,

$$B_{1,sav,1} = 6.77 \times \left(1 - \frac{0.15}{0.3522}\right)$$

$$= 3.887 \text{ tonnes (3 d. p.)}$$

Similarly applying the formulation to determine the biomass savings per stove for each year

y	B_{old} (t)	$\eta_{new,1}$	η_{old}	$B_{y,sav,1}$ (t)
1	6.77	0.3522	0.15	3.887
2	6.77	0.3365	0.15	3.752
3	6.77	0.3208	0.15	3.604
4	6.77	0.3051	0.15	3.441
5	6.77	0.2893	0.15	3.260
6	6.77	0.2736	0.15	3.058
7	6.77	0.2579	0.15	2.832

Finally, to determine the emissions reductions in year y , Equation 1 is employed using the fixed parameters outlined as follows.

Table 10. Fixed parameters and their source to calculate emissions reductions under the methodology

Parameter	Value	Source
B_{old}	6.77 tonnes	Baseline Surveys (for each instance)
$\eta_{new,s}$	36.01%	Manufacturer, Independent laboratory test
η_{old}	15%	Methodology
NCV_{wood_fuel}	0.0156 TJ/tonne	Methodology
EF_{wf,CO_2}	112 tCO ₂ /TJ	Methodology
$EF_{wf,nonCO_2}$	9.46 tCO ₂ /TJ	Methodology
f_{NRB}	0.894	Methodology Tool (see section 6.1)
u_d	26%	Methodology
Adj_{LE}	0.95	Methodology

To provide an example of using Equation 1 with the parameters outlined in Table 10, for $y = 1$

$$ER_{1,1} = 3.8870 \times 100,000 \times 1 \times 1 \times 0.894 \times 0.0156 \times (112 + 9.46) \times 0.95 \times (1 - 0.26) \times 0.5$$

$$= 231,532 \text{ tCO}_2\text{e}$$

An additional factor 0.5 is applied given ICS distribution occurs throughout the year, and this will be applied for other years in line with the ICS distribution ramp-up schedule stated earlier. Note accordingly that a factor 0.75 is to be used specifically for year 3 as distribution is planned only over the first half of the year.

Applying the emissions reduction formulation for the proposed project activity over the first crediting period

y	$B_{y,sav,1} \text{ (t)}$	$N_{y,1}$	$ER_{1,y} \text{ (tCO}_2\text{e)}$	$ER_{2,y} \text{ (tCO}_2\text{e)}$	$ER_{3,y} \text{ (tCO}_2\text{e)}$	$ER_y \text{ (tCO}_2\text{e)}$
1	3.887	100,000	231,532	—	—	231,532
2	3.752	300,000	447,011	463,065	—	910,076
3	3.604	400,000	429,383	894,023	347,299	1,670,705
4	3.441	400,000	409,938	858,767	447,011	1,715,716
5	3.260	400,000	388,378	819,876	429,383	1,637,637
6	3.058	400,000	364,339	776,756	409,938	1,551,033
7	2.832	400,000	337,368	728,678	388,378	1,454,424

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21-Jan-2023 to 31-Dec-2023	206,521	—	—	206,521	—	206,521
01-Jan-2024 to 31-Dec-2024	860,231	—	—	860,231	—	860,231
01-Jan-2025 to 31-Dec-2025	1,646,313	—	—	1,646,313	—	1,646,313
01-Jan-2026 to 31-Dec-2026	1,719,647	—	—	1,719,647	—	1,719,647
01-Jan-2027 to 31-Dec-2027	1,641,975	—	—	1,641,975	—	1,641,975
01-Jan-2028 to 31-Dec-2028	1,555,844	—	—	1,555,844	—	1,555,844
01-Jan-2029 to 31-Dec-2029	1,459,791	—	—	1,459,791	—	1,459,791
01-Jan-2030 to 20-Jan-2030	80,801	—	—	80,801	—	80,801
Total	9,171,123	—	—	9,171,123	—	9,171,123

The estimated annual average GHG emissions reductions over the first crediting period is 1,310,160 tCO₂e.

6. MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	f_{NRB}
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity that can be established as non-renewable biomass
Source of data	Calculated by determining the share of non-renewable woody biomass in the total quantity of woody biomass consumed in the project area. The parameter value is obtained by using CDM Methodological Tool 30²² with inputs specific to the local state of the project instance. A new value is determined every time the project activity is undertaken in a new state of India. For the first instance the parameter value is 0.894. The calculation based on Tool 30 is given in the f_{NRB} spreadsheet provided. Note that an additional uncertainty parameter $u_d = 26\%$ is applied to f_{NRB} in Equation 1 under the methodology (see section 5). From survey results under the first instance it was recorded that beneficiaries primarily sourced firewood from forests which has led to widespread deforestation across the region (as shown for example in Figure 8).
Value applied	0.894
Justification of choice of data or description of measurement methods and procedures applied	The value as per CDM Methodological Tool 30: Calculation of the fraction of non-renewable biomass¹⁸ is calculated using $f_{NRB} = \frac{NRB}{NRB + RB}$ where NRB is the quantity of non-renewable biomass consumed in tonnes and RB is the quantity of renewable biomass in tonnes in the area

²² <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-30-v4.0.pdf>

Purpose of data	Calculation of emissions reductions
Comments	--

Data / Parameter	NCV_{wood_fuel}
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass
Source of data	Value obtained from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2 Energy, Chapter 1 Introduction
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	As per methodology guidelines
Purpose of data	Calculation of emissions reductions
Comments	--

Data / Parameter	EF_{wf,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emissions factor for the use of non-renewable woody biomass in the baseline scenario
Source of data	As per methodology guidelines

Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	--
Purpose of data	Calculation of emissions reductions
Comments	--

Data / Parameter	$EF_{wf,nonCO2}$
Data unit	tCO ₂ e/TJ
Description	Non-CO ₂ emissions factor for the use of non-renewable woody biomass in the baseline scenario
Source of data	As per methodology guidelines
Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	--
Purpose of Data	Calculation of emissions reductions
Comments	--

Data / Parameter	u_d
Data unit	Fraction

Description	Uncertainty deduction for f_{NRB}
Source of data	As per methodology guidelines
Value applied	0.26
Justification of choice of data or description of measurement methods and procedures applied	--
Purpose of data	Calculation of emissions reductions
Comments	--

Data / Parameter	Adj_{LE}
Data unit	Fraction
Description	Factor to account for leakage related to the non-renewable woody biomass saved by the project activity
Source of data	As per AMS-II G methodology guidelines
Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	Discussed in section 5.3 and 5.4
Purpose of Data	Calculation of emissions reductions
Comments	--

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Methodology
Value applied	15%
Justification of choice of data or description of measurement methods and procedures applied	Default value for the baseline device which is a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney, or three-stone fire using firewood (not charcoal)
Purpose of data	Calculation of emissions reductions
Comments	If at any point charcoal is found to be used by baseline devices within an instance, a value of 0.25 will be applied in line with the methodology

Data / Parameter	$\eta_{new,s}$
Data unit	Fraction
Description	Efficiency of the project ICS at the start of the project activity
Source of data	Manufacturer specifications and independent verification from laboratory water boiling tests (WBTs) conducted by the Department of Mechanical Engineering at Jorhat Engineering College (see Appendix 2)
Value applied	36.01%
Justification of choice of data or description of measurement methods and procedures applied	WBTs involve averaging the cold-start and hot-start high-power phases of an ICS ²³
Purpose of data	Calculate emissions reductions

²³ <https://cdm.unfccc.int/methodologies/SSCmethodologies/clarifications/21559>

Comments	The efficiency value corresponds to the Tramontana ICS to be distributed under project activity instances. In case of inclusion of any new ICS models, the corresponding model efficiency will be used for emissions reduction calculation purposes.
Data / Parameter	B_{old}
Data unit	Tonnes per year
Description	Annual quantity of woody biomass which would be used by a household in the absence of the project activity to meet their thermal energy needs
Source of data	Measurement from Baseline surveys
Value applied	For the first project instance in Andhra Pradesh conducted between November and December 2022, an average value of 18.54 kg per day was obtained for each baseline stove, which is equivalent to 6.77 tonnes annually
Justification of choice of data or description of measurement methods and procedures applied	This parameter is determined at the project instance level. The calculation is based on an average of all baseline sample surveys conducted for the respective project instance. Each survey value is determined by first asking each stove user in the sample baseline to physically gather the amount of firewood consumed each day (or over a defined period) on average across all seasons and subsequently having a member of the field distribution team directly take a measurement of the weight using either analogue or digital scales. The value of the parameter is determined once for each project instance and remains fixed for the entire project period. It is calculated after baseline surveys are carried out, but prior to ICS installation given all baseline stoves are aimed to be removed/demolished during the distribution process. Baseline data is gathered extensively and uniformly across a project instance with samples collected from all over the district. Baseline sampling is described in section 3.4.
Purpose of data	Calculation of emissions reductions

Comments

If at any point charcoal is found to be used by baseline devices within an instance, a wood-to-charcoal conversion factor of 4 will be applied in calculations in line with the methodology

6.2 Data and Parameters Monitored

Data / Parameter	$n_{y,i,j}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type i in instance j operating during year y
Source of data	Monitoring surveys
Description of measurement methods and procedures to be applied	The parameter is monitored based on representative sampling of the database of distribution records. As per the methodology, the sampling complies with the latest version of CDM guidelines ²⁴ . Through the surveys beneficiaries are questioned on their ICS usage practices over the monitoring period.
Frequency of monitoring/recording	At least annually
Value applied	Based on results from monitoring surveys Aiming for all ICS distributed to be operational owing to regular maintenance/repair and replacement if necessary
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	A phone number is provided to every end user to contact in case of any issues faced with their ICS, so that the Project Proponent can have them resolved imminently. The project aims to exceed the required sample confidence threshold through using techniques such as oversampling
Purpose of data	Calculation of emissions reductions
Calculation method	Based on sample-based monitoring, the parameter is calculated as $n_{y,i,j} = \frac{n_{op,y,i,j}}{n_{tot,y,i,j}}$ where

²⁴ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

	n = number of ICS based on sample op means operational, tot means total
Comments	--
Data / Parameter	$N_{0,i,j}$
Data unit	Number
Description	Total number of project devices of type i in instance j commissioned over the period
Source of data	Distribution database
Description of measurement methods and procedures to be applied	Distribution records are created through surveys conducted when each ICS is successfully distributed to a project beneficiary This parameter is not calculated on a sample basis, but instead on the total project population. Thus it does not form part of sample-based monitoring in section 6.3.
Frequency of monitoring/recording	At the time of commissioning/distribution of ICS
Value applied	Sum of the entries in the distribution database over the period
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	During distribution an extensive dataset is obtained on each beneficiary, including personal details, ID card/community certificate information and photo, and an individual photo together with their new ICS
Purpose of data	Calculation of emissions reductions
Calculation method	--
Comments	--
Data / Parameter	n_{HH}

Data unit	Number
Description	Number of ICS distributed to each household
Source of data	Distribution database
Description of measurement methods and procedures to be applied	When an ICS is distributed by a field team member, the QR code is scanned and a corresponding family record is created and synced into the distribution database. Each family record is associated with a unique BPL card (see section 1.14). When there are multiple ICS records linked to a single BPL card, it signifies that multiple ICS have been distributed to a single family. This parameter is not calculated on a sample basis, but instead on the total project population. Thus it does not form part of sample-based monitoring in section 6.3.
Frequency of monitoring/recording	At the time of commissioning/distribution of ICS
Value applied	1 as standard. If a family requests more than one ICS, the Project Proponent will consider providing an additional ICS if, amongst other additional details obtained, the amount of firewood used is measured to be significantly higher than the average across the mandal.
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	--
Purpose of data	Calculation of emissions reductions
Calculation method	--
Comments	--

Data / Parameter	$\eta_{new,y,i}$
Data unit	Fraction
Description	Thermal efficiency of ICS of type i in year y

Source of data	Water boiling tests (WBTs) or equivalent as part of monitoring
Description of measurement methods and procedures to be applied	The parameter is measured by conducting WBTs involving averaging the cold-start and hot-start high-power phases of an ICS. In line with the methodology, for an ICS of type i and vintage y a sample from the first ICS batch of 3 ICS are selected and for each one 3 WBTs are conducted. The degradation of efficiency measured from monitoring the first ICS batch may be applied across all batches. The WBTs will be conducted by an independent technical laboratory and certified test results will be provided as part of monitoring.
Frequency of monitoring/recording	Annually
Value applied	Based on an average of the results from the WBTs
Monitoring equipment	Laboratory tests
QA/QC procedures to be applied	Experienced technical laboratories are selected for the purposes of conducting all WBTs The test results from the 9 samples must meet the standard deviations limit and 90/10 confidence/precision as set by the methodology, else more sample tests will be undertaken until the requirement is met. This is discussed further in section 6.3.
Purpose of data	Calculation of emissions reductions
Calculation method	For an ICS of type i and vintage y, the parameter is determined by averaging the results from independent WBTs conducted on a sample of 3 operational ICS
Comments	--

Data / Parameter	μ_y
Data unit	Fraction
Description	Stove usage adjustment factor accounting for any continued use of pre-project or other devices

Source of data	Monitoring surveys
Description of measurement methods and procedures to be applied	As the B_{old} method is used in emissions reduction calculations, this parameter forms a part of the methodology (Equation 7 of AMS-II G). During distribution the signed end user agreement includes provisions to remove or demolish the pre-project stoves from households eligible for a new ICS. This is expected to be enforced and therefore the value for this parameter is expected to be 1. To check for any pre-project or other devices being used, sample-based monitoring will be applied as described in section 6.3. If it is found that a non-project stove is being used following conducting a monitoring survey, the parameter μ_y is calculated for the corresponding user as the ratio of the frequency of usage of the project ICS compared to the frequency with which all stoves are used by the household over a defined time period. For instance, if it is found from surveys that a non-project stove is being used 7 times per week and the ICS is being used 21 times a week, the following calculation would be applied $\mu_y = 21 / (21 + 7) = 0.75$
Frequency of monitoring/recording	Expected annually but at least biennially
Value applied	1 unless pre-project or other devices are found to be used
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	--
Purpose of data	Calculation of emissions reductions
Calculation method	A parameter value of one if only project ICS are being used, otherwise apply the follow calculation over a defined time period $\mu_y = \frac{F}{F + f}$

	where f = frequency of non-project stove usage F = frequency of project ICS usage
Comments	--

Data / Parameter	<i>Life Span</i>
Data unit	Number of years
Description	Operating lifetime of project ICS
Source of data	Manufacturer specifications
Description of measurement methods and procedures to be applied	All ICS are manufactured to a fixed specification/error tolerance (see Table 4) and quality control checks are undertaken by the field teams prior to distribution to ensure that the ICS have standardised dimensions and no damage. A one-year warranty is also provided by the ICS manufacturer.
Frequency of monitoring/recording	Once at project installation
Value applied	7
Monitoring equipment	--
QA/QC procedures to be applied	--
Purpose of data	--
Calculation method	--
Comments	It is considered that the project device may be in operation beyond 7 years with proper maintenance and this will be checked and reported during monitoring. If the lifespan of a project device ends up being shorter than 7 years, either it will be proven that the device has been replaced, or if no replacement has occurred there will be no emissions reductions claimed

Data / Parameter	<i>Date of commissioning of project device i and the corresponding instance j</i>
Data unit	Date and instance number
Description	Exact date of installation at household of each individual ICS and the project instance to which it belongs
Source of data	Distribution database
Description of measurement methods and procedures to be applied	--
Frequency of monitoring/recording	Once at ICS installation
Value applied	--
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	--
Purpose of data	Calculation of emissions reductions
Calculation method	--
Comments	--

6.3 Monitoring Plan

A sampling plan is required for the purposes of monitoring and verification of project emissions reductions. As per CDM guidelines ²⁵ a sampling plan specifically contains (i) a sample design/sizing and field data and (ii) an implementation plan and data quality control.

²⁵ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

6.3.1 Sample design/sizing and Field data

The target population includes all households under the grouped project who are having their traditional stoves replaced by a new Tramontana ICS. Owing to the large number of ICS planned to be distributed under the project, representative sampling of the population is applied, in line with the CDM sampling guidelines.

The objective of sampling is to obtain an unbiased statistically significant estimate of the lifecycle variables to be monitored from the population, presented in Table 11, for the purposes of emissions reduction calculations over the course of the project period. The parameter type column specified in the table corresponds to either proportion or mean value as per the methodology guidelines.

The parameter $\eta_{new,y,i}$ in Table 11 is monitored using a separate approach outlined in the methodology. More specifically, an average is calculated from 9 samples undertaken (3 ICS with 3 WBTs conducted for each), with sampling required to meet the standard deviations limit of 0.05 and fit the 90/10 confidence/precision level. If the requirement is not met, more samples will be taken to achieve this. The simple random sampling approach described below is applied to identify the specific ICS to be tested for the respective monitoring period. Note that samples will be taken entirely from the first batch of an ICS type, and the corresponding efficiency values obtained for each ICS vintage will be applied for all subsequent project ICS distributed of that type. The remainder of this subsection will focus on sampling methods to determine the first two parameters in Table 11.

Sampling is conducted separately and independently for each project activity instance under the grouped project for the first two parameters in Table 11, to improve data reliability and accuracy. The sampling frame is taken as the set of all information recorded in the online database for a project instance.

As per CDM standards²⁶, a 90/10 confidence/precision is to be used for all small-scale projects and 95/10 for all large-scale projects, and in line with methodology AMS-II G a 90/10 confidence/precision when undertaking annual sampling and 95/10 when adopting biennial sampling inspections. We apply a minimum 95/10 confidence/precision when sampling the parameters for monitoring purposes.

²⁶ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

Table 11. Parameters which are sampled and monitored through the project lifecycle

Parameter	Type
$n_{y,i,j}$ Proportion of project devices of type i in instance j operating during year y	Proportion
μ_y Stove usage adjustment factor accounting for any use of pre-project or other stoves during year y	Proportion
$\eta_{new,y,i}$ Thermal efficiency of ICS of type i in year y	Mean Value

There is no necessity to account for different ICS vintages when sampling, given distribution for a single project instance is unlikely to continue to occur over many years. It should also be noted that geographical variations have minimal influence on the monitored parameters within an instance. Therefore owing to the homogenous nature of an instance population, a Simple Random Sampling technique is used for sampling.

Note that if for any reason a new ICS model or variation on the existing ICS model is included into the project in the future, it will be grouped into a separate population unit and the sampling approach described in this section will be applied to it.

To undertake a Simple Random Sampling approach, a random number generator must be utilised. Given each ICS has a unique serial code identifier, we can first pair each ICS with a sample number. For instance, the samples can begin from 1 and increase up to the total population size of the instance corresponding to all successfully distributed ICS. The random number generator can then be applied to randomly select an ICS from the population. The task is repeated until the targeted sample size is achieved.

Once all ICS in the sample have been identified and mapped, the first two monitored parameters in Table 11 are determined through field surveys conducted by the team. More specifically, for each of the parameters:

$n_{y,i,j}$ A site visit is carried out to inspect the household and identify whether the ICS is in operation. A detailed survey is also conducted with the end user to identify

current usage patterns and operation/performance over the past year.

μ_y A site visit is carried out to inspect the household and identify whether any pre-project/other stoves are in operation. A detailed survey is also conducted with the end user to check whether their traditional stove was removed/demolished and if it or any other stove has been used.

In addition to covering these monitored parameters, monitoring surveys provide a wider and more comprehensive review of recent ICS usage, performance and perceptions by end users.

6.3.1.1 Sample size

To determine the sample size for the first two parameters in Table 11 required to represent the instance population, we consider Simple Random Sampling mentioned earlier, which can be formulated as

$$n \geq \frac{1.96^2 N p (1 - p)}{0.1^2 p^2 (N - 1) + 1.96^2 p (1 - p)} \quad \text{Equation 10}$$

where

n = Sample size
 N = Population size (total number of households or ICS in instance)
 p = Expected proportion for the sampled parameter
1.96 = Confidence level (equivalent to 95%)
0.1 = Relative precision level

The 95% confidence level is used with a precision or margin of error in the estimate that is not more than $\pm 10\%$ in relative terms to the value of the proportion parameter. For this method, the proportion p corresponding to the expected value of the sampled parameter must be estimated. As per CDM guidelines²⁷, it can be estimated either by

- (i) referring to the results of previous studies and using these results
- (ii) in a situation where information from previous studies is not available, taking a preliminary sample as a pilot and using that sample to provide the estimates, or
- (iii) using best guesses based on the researcher's own experiences

For registration purposes option (iii) will be used. Thus we select a value of $p = 0.8$, which is conservative for the first two monitored parameters in Table 11 as the Project Proponent is continuously repairing/replacing all ICS and all traditional cookstoves are aimed to be demolished/removed. Note that for the parameter μ_y , the expected proportion p here in the sample size calculation corresponds to the proportion of traditional (or other) stoves that are *not* being used (i.e. a failure rate²⁷). For subsequent monitoring periods, option (i) above could

²⁷ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

alternatively be followed to estimate p by using a value determined from the previous monitoring period.

The Simple Random Sampling approach outlined will be applied for sampling each of the parameters $n_{y,i,j}$ and μ_y . Once the sample size is obtained for each parameter, the greater value is selected in order to be able to combine monitoring for each period into a single survey. Conducting a single survey allows for optimisation from a time and cost perspective and is a more conservative approach.

In the following we present an example application of the sample sizing approach described above.

Taking the case of the first project activity instance where the ICS is distributed in the ASR district of Andhra Pradesh, it is estimated that the total number of beneficiary households (population N) will be around 50,000. In addition, a value of $p = 0.8$ for μ_y and $p = 0.75$ for $n_{y,i,j}$ is taken for illustrative purposes.

Applying Equation 10 denoting Simple Random Sampling, first for parameter u_y , we calculate the sample size for a 95/10 confidence level/precision as

$$n \geq \frac{1.96^2 \times 50,000 \times 0.8(1 - 0.8)}{0.1^2 \times 0.8^2 \times (50,000 - 1) + 1.96^2 \times 0.8(1 - 0.8)}$$

$$n \geq 95.86$$

$$n = 96$$

Similarly, applying the formulation for parameter $n_{y,i,j}$, we obtain

$$n \geq \frac{1.96^2 \times 50,000 \times 0.75(1 - 0.75)}{0.1^2 \times 0.75^2 \times (50,000 - 1) + 1.96^2 \times 0.75(1 - 0.75)}$$

$$n \geq 127.73$$

$$n = 128$$

To be able to combine both μ_y and $N_{y,i,j}$ into a single monitoring survey as planned, one then takes the larger of the two outputs above, represented by 128, as the minimum size threshold for sampling over the respective monitoring period.

6.3.2 Implementation plan & Data quality control

The Project Proponent will oversee and manage the monitoring process and sampling plan to ensure accurate measurement and reporting of emissions reductions from the project activity throughout the project life. The methods set out in Table 12 are applied for the purposes of monitoring the parameters from Table 11, and this is undertaken with representative sampling as outlined in section 6.3.1 to identify the specific households to be included in the respective monitoring cycle.

Table 12. Frequency and methods with which project parameters are monitored

Parameter	Monitoring frequency	Frequency as required by methodology	Monitoring methods to be applied	Seasonal fluctuation
$n_{y,i,j}$	Expected annually and will occur at least biennially	Biennial	Site visits to undertake visual examination and conducting detailed surveys/interviews	None
μ_y	Expected annually and will occur at least biennially	Biennial	Site visits to undertake visual examination and conducting detailed surveys/interviews	None
$\eta_{new,y,i}$	Expected annually	Annual	WBTs conducted by independent technical laboratories	None

Substantial oversampling is applied for the first two parameters in Table 12 in order to enhance the reliability of the data recorded, as well as to adjust for any errors or cases of answer bias that may be observed. Note that outliers do not exist for these monitoring parameters as they correspond to proportions as opposed to mean values under the methodology.

To successfully implement the project, an extensive training programme is conducted prior to distribution for each instance for all local field staff and coordinators employed for the purposes of data collection. This includes classroom lectures (as shown in Figure 17) outlining survey structure, purpose and goals, demonstrations of the Tramontana ICS software application (see Figure 5), as well as on-field practice sessions to gain experience in using the tablets containing the application. A key additional aspect of the training programme is to convey engagement techniques with the community, approaches to minimising errors, and methods to deal with issues faced such as non-response cases, conflicts of interest and interviewer bias. Members of the local community are typically employed as field staff in order to maintain closer community engagement and maximise response rate, whilst it also provides support to the local people in the form of employment and training. All field staff within a mandal are accompanied by an experienced mandal coordinator who helps oversee distribution, ensuring the team works in an efficient manner and to help tackle any problems faced.



Figure 17. An internal classroom training session held for field team members in advance of ICS distribution

All surveys used for the project have been designed and tested by the Project Proponent in consultation with the various stakeholders. The surveys have been programmed into the Tramontana ICS software application which is used by the field teams. The team are able to view the application on tablets, which have been purchased specifically for the purposes of the project activity. The application has been developed to work in an offline mode given the remoteness of project households/beneficiaries. Once a field team member has successfully completed a survey (after meeting all requirements and passing the built-in checks in the application), they can submit the entry on the tablet to create a record for the respective beneficiary. The built-in checks in the application such as mandatory fields/data types minimise any incorrect or incomplete surveys from being submitted. The submitted record is stored locally and will be automatically synced to a Google cloud database once the tablet is brought online under an internet connection. This automation of syncing maintains the data integrity and minimises any risk of data loss, theft, tampering/fraud or negligence/errors in translation. Once the data is in the cloud database, it is automatically processed and translated into an Excel sheet format which can be easily analysed by the VVB, Verra and any other project parties.

Once distribution/monitoring data has been recorded, submitted and translated into the Excel sheet, it is checked by a dedicated office team for the purposes of quality control against a pre-defined set of criteria developed by the Project Proponent. The team's tasks include checking all: written responses, photos (including their content, clarity and positioning), signed licence agreements and location coordinates. Any deviation from the set of criteria will be marked, rejected and sent immediately for resurveys in the field. Incorrect surveys are later removed from the database once the resurveys have been successfully completed. Multiple tiers of checking are undertaken by the office teams as part of the quality control process to ensure all incorrect entries are identified.

Note that at the initial stage after ICS distribution some additional checks are also carried out. This includes a physical household visit to all ICS beneficiaries in the weeks following distribution, to ensure the ICS is being used correctly, to provide helpful tips (e.g. optimal processing methods for the firewood) and to obtain feedback on the ICS performance to date. Simultaneously, a separate dedicated team focusses on calling and checking all phone numbers in the database that were provided by households during distribution, to ensure beneficiaries are reachable if needed at any time by the Project Proponent.

Any ICS that is found not to be in use post-distribution or from a monitoring survey will not be counted towards the total number of ICS in operation for the preceding period or thereafter. Further, any use of alternative stoves, for instance which may have been built or purchased since the distribution of the project ICS, will be accounted for using the method described in the table in section 6.2 for the monitored parameter μ_y .

The final data available in the online database will include (i) initial information obtained during ICS distribution (outlined in section 1.12.1) and (ii) monitoring data recorded from sample monitoring surveys (and laboratory tests) as outlined in section 6.2. The data will be extensive and directly attributable to a particular ICS/household, forming clear evidence for any emissions reductions that are claimed.

For the purposes of monitoring, the Project Proponent will compile a regular monitoring report to be submitted to the VVB as part of project audits throughout the project life. This will detail the monitoring activity and monitoring data recorded in the online project database and from laboratory tests, as well as the associated calculations specifying the emissions reductions from the project activity for the respective period.

7. QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

The procedures detailed in section 6 are followed for monitoring for the first verification period between 21 January 2023 and 23 August 2023.

As part of monitoring, 840 randomly sampled surveys have been conducted, which is several multiples higher than the 96 minimum requirement from sample size calculations (see datasheet provided). In addition a more extensive set of questions are asked in the surveys around ICS benefits, fuel use purpose, any issues faced and taking photos of all users with their ICS. All surveys were completed using the Tramontana application on tablets by team members employed from the local community.

All baseline, distribution and monitoring databases for the first verification period are provided to the auditor and certification body Verra.

7.1 Data and Parameters Monitored

Data / Parameter	$n_{y,i,j}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type i in instance j operating during year y
Value applied	From the monitoring survey sample results for the first period, it was observed that all ICS distributed were being used by the beneficiaries 7 days a week. $n_{y,1,1} = \frac{n_{op,y,1,1}}{n_{tot,y,1,1}} = \frac{840}{840} = 1$ where n = number of ICS based on sample op means operational, tot means total
Comments	100% of the beneficiaries stated that the functioning of their ICS as 'excellent' and having experienced no breakdowns so far. All beneficiaries also confirmed that regular maintenance/repair and replacement services were being offered. A phone number has been provided as well as access to a local representative in case of any issues.

Data / Parameter	$N_{0,i,j}$
Data unit	Number
Description	Total number of project devices of type i in instance j commissioned over the period
Value applied	From distribution database records for the first period, it was observed that $N_{0,1,1} = 46160$ As part of the database, records of the beneficiary were captured including personal details, ID card/community certificate information and photo, and photo of the user together with their new ICS
Comments	Currently all ICS distributed are of the same type, presented in section 1.12, and are part of the first project instance.

Data / Parameter	n_{HH}
Data unit	Number
Description	Number of ICS distributed to each household
Value applied	From distribution database records, it was observed that there were no BPL card duplicates found. Therefore all households under the instance only received a single ICS, with $n_{HH} = 1$
Comments	Over the monitoring period, some beneficiaries commented that having access to two (or a double pot) ICS would be useful. This feedback is being evaluated by the Project Proponent and will be discussed with the communities further.

Data / Parameter	μ_y
Data unit	Fraction
Description	Stove usage adjustment factor accounting for any continued use of pre-project or other devices
Value applied	Based on the monitoring survey sample results for the first period, it was observed and explicitly confirmed by all beneficiaries that their pre-project stove had been removed or demolished, and also confirmed that no one was using their original stove. Therefore this parameter takes a value of 1.
Comments	The feedback obtained on the ICS received was overwhelmingly positive, as seen in the monitoring database. It was also explicitly stated that all sampled beneficiaries would continue to use their ICS in the future. There has been no change in the type of fuel used by the ICS, with all beneficiaries continuing to use firewood which they had been using in the pre-project scenario.

Data / Parameter	<i>Life Span</i>
Data unit	Number of years
Description	Operating lifetime of project ICS
Value applied	7
Comments	Quality control checks were undertaken at the warehouses by field team members prior to distribution of the ICS. Any damaged or non-standard parts were sent back for replacement. All project devices are observed to be in operation and none were observed to have reached their end of life as of yet.

Data / Parameter	<i>Date of commissioning of project device i and the corresponding instance j</i>
Data unit	Date and instance number
Description	Date of installation at household of each individual ICS and the project instance to which it belongs
Value applied	Synced date/time and instance number are recorded in the monitoring database provided
Comments	The synced date and time of ICS installations into the monitoring database is automatically recorded through the Tramontana software application and used for this parameter

The thermal efficiency $\eta_{new,y,i}$ of the ICS denoted in section 6.2 is monitored with an annual frequency and therefore not included in this monitoring period.

The SDGs described in Table 5 have been monitored during the monitoring period and the results are presented in Table 6 in section 1.18. More specific details can be found in the baseline, distribution and monitoring databases provided.

7.2 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net emissions reductions achieved by the project as highlighted in section 5.4.

7.3 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emissions reductions achieved by the project as highlighted in section 5.4.

7.4 Leakage Emissions

Leakage is estimated using the parameter Adj_{LE} in section 6 in accordance with the AMS-II G methodology. The parameter value of 0.95 is conservative in reality, given across the region there was found to be no evidence of non-project users utilising the firewood saved under the project, as well as little incentive to do so given the plentiful nature and zero cost of firewood.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

The values in the table below are taken from the totals calculated in the monitoring database.

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)
21-January-2023 to 23-August-2023	--	--	--	62,743	62,743
Total	--	--	--	62,743	62,743

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
21-January-2023 to 23-August-2023	79,589	62,743	21.2%	The percentage difference in emissions reductions is owing to the monsoon weather between June and August which caused some delays in ICS distribution. In addition, much time had to be spent conducting resurveys due to errors observed in the database data recorded by the field team, which is required to meet the pre-defined high standards set by the Project Proponent. As the team gain experience the number of resurveys will reduce. 46,160 ICS were distributed over the first monitoring period as opposed to an estimated 58,630 by this stage.

APPENDIX 1

The schedule followed including the event details for stakeholder meetings conducted for the first project instance in Andhra Pradesh is outlined.

Mandal	Village	Meeting Venue	Attendees	Date	Special Guests
Paderu	Modapalli	AMC Yard	105	14-Nov-2022	Kottagulli Usha Rani (President (Sarpanch) of GP Paderu) Sonari Ratnakumari (Mandal Parishad President of Paderu) Sonari Venkata Ramana Murthy (ex-Mandal Parishad President of Paderu)
	Onuru	Anganwadi Centre	52	15-Nov-2022	Thokala Eswara Rao (MPTC of Vantalamamidi) Pangi Rambabu (President of Vantalamamidi) Palliki Lakku (CFFI National Member)
	Paderu	Sri Modhakonda mmathalli Temple	310	28-Dec-2022	Sri V. Abhishek (Sub Collector of A.S.R. district) Sri Shathakabulli Babu (Tricor Chairman of Paderu) Kotagulli Usharani (President of Paderu) Rathnakumari (MPP of Paderu) Sri Ramana Murthy (Former MPP of Paderu)
G. Madugula	G. Madugula	Rachabanda	108	16-Nov-2022	Mathyarasa Gayathri Devi (Market Yard Chairman of G. Madugula and Paderu mandals) Kimudu Rambabu (President of G. Madugula)

					Mathyasara Vijaya Kumari (MPTC Member of G. Madugula) Pangi Chitti Babu (MPTC of G. Madugula)
	Gaduthuru	Anganwadi	137	17-Nov-2022	Kondapalli Simhachalam (President of Gaduthuru) Madhusudhan Babu (Secretary of Gaduthuru GP) K. Chandrapathi (President of PESA (Panchayaths Extension to Scheduled Area) of Gaduthuru)
Hukumpeta	Dabavalasa	Jio Tower	79	20-Nov-2022	Korra Raja Rao (Former MPTC of Hukumpeta) Kincheyi Krishna (President of Dabavalasa)
Pedabayalu	Galaganda	Panchayath Office	82	19-Nov-2022	Pujari Lohidasu (President of Galaganda) Thalla Bullamma (Former President of Galaganda) Marri Devayya (Former President of Pedakodapalli)
	Pedabayalu	Girijana Bhavan	63	21-Nov-2022	Korra Raju Babu (Vice MPP of Pedabayalu) Vanthala Ananda Rao (MPTC of Pedabayalu) Gullili Krishna Rao (President of Pedabayalu)
Chinthapalli	Gondhipakalu	Grama Sachivalayam	102	22-Nov-2022	Thambeli Mohan Rao (MPTC of Chinthapalli) Sagila Varalaxmi (President of Gondhipakalu)
	Yerrabandha	GCC Godowns	103	24-Nov-2022	Segge Sathibabu (MPTC of Chinthapalli)

					Lotha Pandayya (President of Pedhuru) Kuda Chinnabbai (Former President of Pedhuru) Kuda Raju Babu (VRP NREGS of Pedhuru)
G.K. Veedhi	Devarapalli	Grama Sachivalayam	118	23-Nov-2022	Pothuraju Krishna Murthy (MPTC of Devarapalli) Siribala Bujji Babu (President of Devarapalli) Matchala Mathya Raju (Former ZPTC of Devarapalli)
	Kothapalem	MPP School	102	25-Nov-2022	Motanam Sathyanarayana (MPTC of G.K. Veedhi) Kankipati Veerendra Prasad (President of Kothapalem village) Kankipati Ramarao (Former MPTC of G.K Veedhi)
Araku Valley	Baski	Church Ground	107	26-Nov-2022	Ranjapalli Usharani (MPP of Araku Valley) Padi Ramesh (President of Baski) Killo Ramanna (Vice MPP of Baski)
	Nandi Valasa	MPP School Building	84	28-Nov-2022	Swabi Ramachandra (MPTC of Madhala) J. Kiran Kumar (Panchayath Secretary of Madhala village)
Ananthagiri	Venaganda	MPP School Building	53	27-Nov-2022	Sembi Sanyasi Rao (President of Vaalasi) Sodipalli Seema (President of Venaganda)

					Siveri Kondala Rao (Former President of Venaganda)
Munchingiputtu	Munchingiputtu	Community Hall	161	30-Nov-2022	Jallipalli Subhadra (Z.P chairperson of Munchingiputtu) P.Sathyanarayana (MPP of Munchingiputtu) K.Kiran Kumar (MPDO of Munchingiputtu) Seesa Subhash (President of Kinchayiputtu)
	Maddulabanda	MP Primary School	65	02-Dec-2022	K. Mukamma (President of Kamada GP) K. Radhakrishna (G.P.S School Teacher in Kamada GP)
Dumbriguda	Dumbriguda	IKP Office	197	01-Dec-2022	V. Nagesh (MPDO of Dumbriguda) M. Geetha (MPTC of Dumbriguda)
	Parasilla	Rachhabanda	75	03-Dec-2022	Shobha Raghupathi (MPTC of Parasilla) Veeraboyina Subbarao (President of Parasilla)
Koyyuru	Yerri Naidu Pakalu	Near Community Hall	65	03-Dec-2022	Badugu Ramesh Babu (MPP of Koyyuru) Jampa Rajakumari (President of Koyyuru)
	Palakajeedi	Government School Building	63	05-Dec-2022	B. Ramesh Babu (MPP of Koyyuru) Dadala Ramesh (President of Palakajeedi) Thammarbha Madhava Rao (MPTC of Palakajeedi)

APPENDIX 2

Laboratory water boiling test (WBT) results on the Tramontana ICS (detailed in Table 3 and Table 4) conducted by the Department of Mechanical Engineering at Jorhat Engineering College

Thermodynamic Laboratory

Department of Mechanical Engineering

Jorhat Engineering College

Jorhat-7, Assam

www.jecassam.ac.in/



1. Water Boiling test results of FUELNZEL (CRS-23) stove

Fuelnzal (CRS-23), a natural draft biomass cook stove was tested in the Thermodynamic Laboratory of the department of Mechanical Engineering, Jorhat Engineering College, Jorhat. The stove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of FUELNZEL (CRS-23) Stove

Specification

Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Net wet	11 kg
Draft	Natural draft

Manufactured by

Ravi Engineering and Chemical Works

ADD:L-166, Sector-3

DSIIDC BAWANA, DELHI 110039

Email: recwfire@gmail.com

1.2 Test results

The average thermal efficiency of the stove was found to be 36%. The summary of the test results is given in the table below.

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil	min	39	37	37	37.7
Temp-corrected time to boil	min	38	36	36	37
Burning rate	g/min	9	10	10	10
Thermal efficiency	%	35%	35%	35%	35.1%
Specific fuel consumption	g/liter	77	77	75	76
Temp-corrected specific consumption	g/liter	74	75	74	74

Temp-corrected specific energy cons.	kJ/liter	1,318	1,330	1,315	1,321
Firepower	watts	2,752	2,899	2,844	2,831
2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot	min	34	37	36	35.7
Temp-corrected time to boil	min	33	36	36	35
Burning rate	g/min	11	9	9	10
Thermal efficiency	%	37.3%	36.9%	37%	37%
Specific fuel consumption	g/liter	77	73	71	73
Temp-corrected specific consumption	g/liter	75	71	71	72
Temp-corrected specific energy cons.	kJ/liter	,332	1,260	1,254	1282
Firepower	watts	3,118	2,743	2,770	2,877
3. LOW POWER (SIMMER)	units	Test 1	Test 2	Test 3	Average
BENCHMARK VALUES (for 5L)		Test 1	Test 2	Test 3	Average
Fuel Use Benchmark Value	g	373	364	361	366
Energy Use Benchmark Value	kJ	6,625	6,475	6,421	6,507
IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	36.4%	35.8%	36.0%	36.01%


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